

Research Methodology and IPR

Course Teacher

Dr. Basavaraj S.Kudachimath

Course Outline

Course Code	22MBA23	CIE Marks	50
Teaching Hours/Week (L:P:SDA)	4:0:0	SEE Marks	50
Total Hours of Pedagogy	50	Total Marks	100
Credits	04	Exam Hours	03

Course Learning objectives:

- To understand the basic components of research design
- To Gain an insight into the applications of research methods
- To equip students with various research analytical tools used in business research
- To provide the insights of IPR and IPR system in India

Unit-1

Introduction to Business Research:

Presentation by
Dr. Basavaraj S.Kudachimath



Discussion Points

- Meaning of Business Research
- Types of Business Research
- Process of research-
 - Defining the research problem,
 - Formulating the research Hypothesis,
 - Developing the research proposals,
 - Research design formulation,
 - Management problem,
 - Sampling design,
 - Planning and collecting the data for research,
 - Data analysis and interpretation.
- Research Application in business decisions,
- Ethical issues in business research.
- Features of a good research study.

What is research?

Research is a process of investigation. An examination of a subject from different points of view. It's not just a trip to the library to pick up a stack of materials, or picking the first five hits from a computer search.

Research is a hunt for the truth. It is getting to know a subject by reading up on it, reflecting, playing with the ideas, choosing the areas that interest you and following up on them.

Research is the way you educate yourself.



Research

What is Research?

Webster's New International Dictionary defined research as, 'careful or critical inquiry or examination in seeking facts or principles; diligent investigation in order to ascertain something.'

According to Clifford Woody 'Research comprises defining and redefining problems, formulating hypothesis or suggested solutions; collecting, organising and evaluating data; making deductions and reaching conclusions; and at last carefully testing the conclusions to determine whether they fit the formulating hypothesis.

D. Slessinger and M. Stephenson in the Encyclopedia of Social Sciences define research as 'the manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether that knowledge aids in construction of theory or in the practice of an art.



Research

Therefore, research could be understood as an organized activity with specific focus [objectives] on a problem or issue supported by compilation of related data and facts, involving application of relevant tools of analysis and deriving logically sound inferences based on originality.

Characteristics of Research

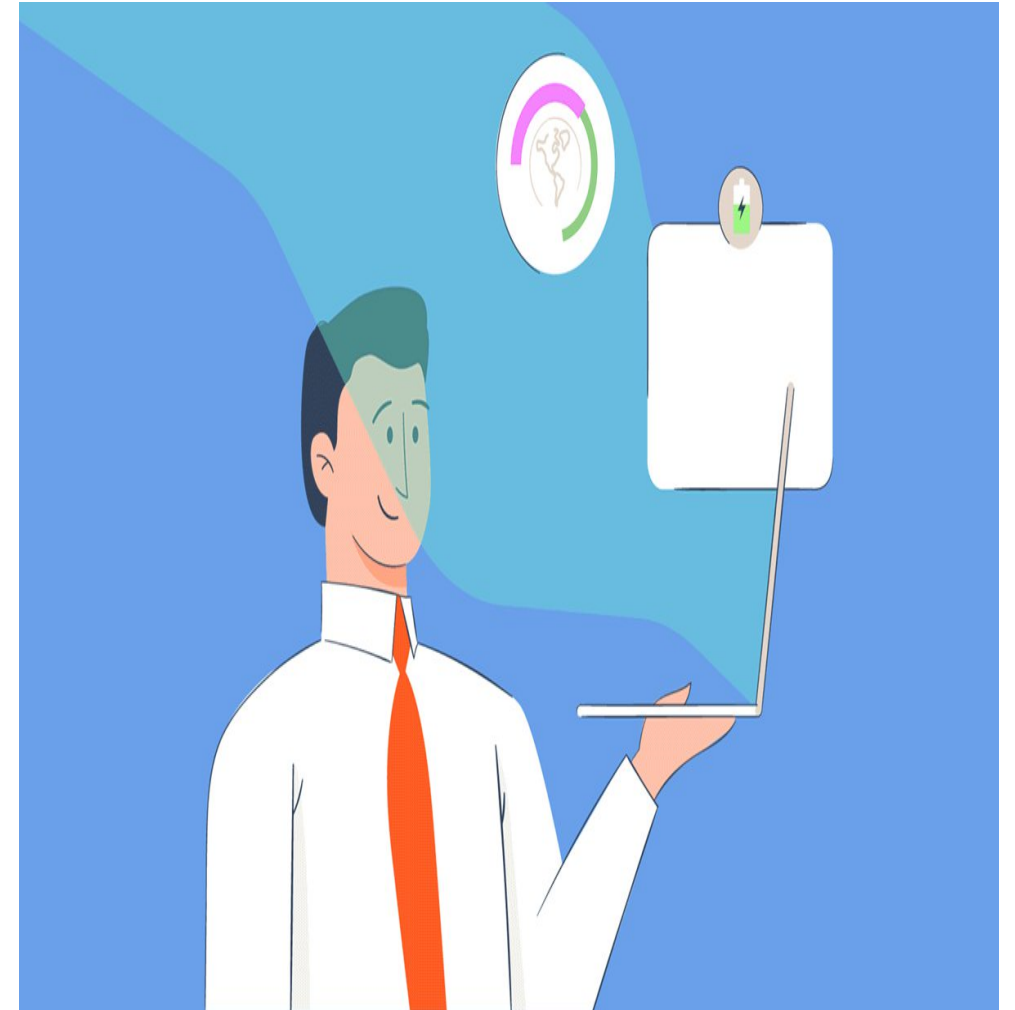
- Systematic Process
- Valid and Verifiable
- Cyclical Nature
- Critical
- Generalization

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Research, at its core, is a systematic and careful investigation into a subject of study to discover new facts or information.

How Business Research defined?

- *According to McDaniel and Gates :*
- "Business research is the planning, collection and analysis of data relevant to business decision-making and the communication of the results of this analysis to management".
- *According to Zikmund :*
- "Business research is a management tool that companies use to reduce uncertainty. It is a manager's source of information about organizational and environmental conditions and covers topics ranging from long range planning to the most ephemeral tactical decisions".
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OBJECTIVES OF BUSINESS RESERARCH(1)

- Promote better decision making - by enough analysis
- Research is the basis of innovation
- Research identifies the problem areas
- Research helps in forecasting
- Research helps in formulation of policies and strategies
- Research helps in development of new products or modifying existing products

Motivation in Research

1. Desire to get research degree along with its simultaneous benefits.
2. Desire to solve the challenging and unsolved problems.
3. Joy of doing some creative work.
4. Desire to give service to society.
5. Desire to get respectability.

Importance of Business Research

1. Business Research helps to solve the Business problem
2. Business Research provides various data for Business decision making
3. Helps to study the business phenomena
4. Govt. needs research reports for planning and execution
5. Research is needed to develop the existing body of knowledge.
6. Research is needed for developing the society in general.

Types of Business Research

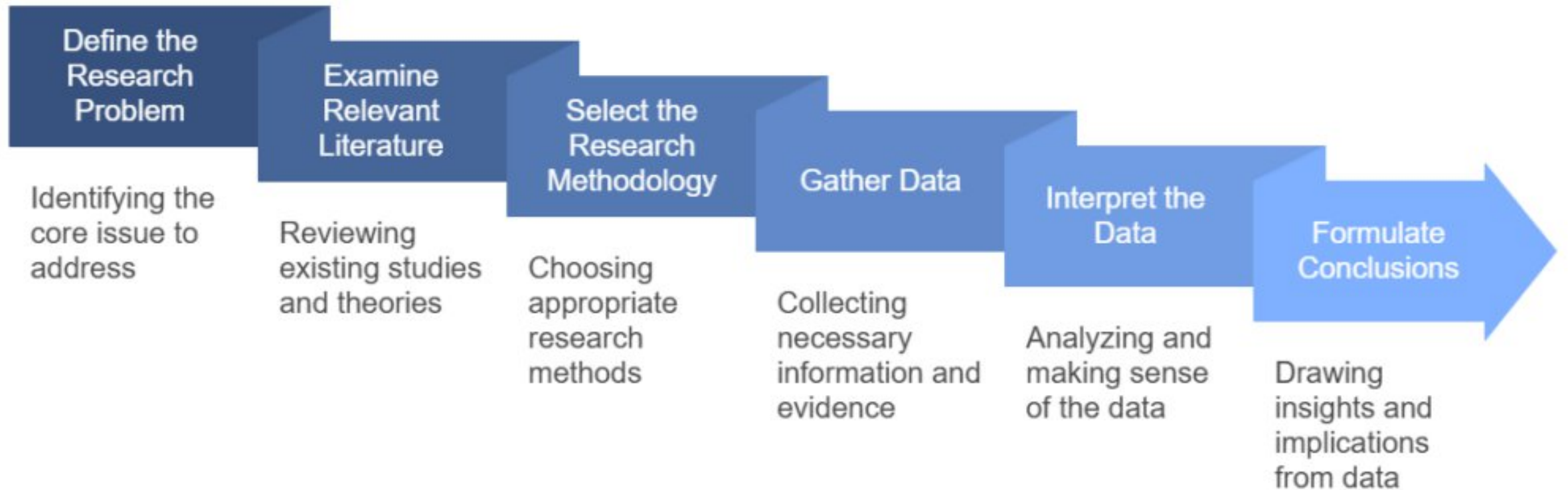
- 1.Descriptive Research. (A research which describes the problem. What? and How?)
- 2.Applied Research. (A research conducted in an organization to find the solution for the problem. (How to solve problem))
3. Fundamental Research. (Research conducted to generalize the phenomena or the tendency (Is it applicable everywhere?)
4. Quantitative research. (Research conducted for variables expressed in terms of quantity)
5. Qualitative research. (Research conducted for variables expressed in terms of quality. (How many?)
6. Conceptual Research. (Research conducted on certain theory or concept)

Types of Research continued

- 7. Empirical Research. (Research conducted on experiments)
- 8. Historical Research. (Research conducted on historical or past data)
- 9. Case study research. (Research conducted on a single unit)
- 10. Scientific Research. (Research conducted in the field of science)
- 11. Social Research. (Research conducted in the field of social behaviors)

Research Process

Business Research Process



Management Vs Research Problem

Management Problem



- Focuses on actions
- Focuses on symptoms
- Asks what the decision maker needs to do.

VS



Research Problem



- Data-oriented
- Focuses on causes
- Asks what information is needed.

1. Defining the Problem

Defining the problem refers to expressing an exact problem statement. In business, management problem has to be redefined as research problem which helps to frame the research work.

Example: Management Problem: Increasing the sales of the company.

Research problem is : Analysing the consumers buying behaviour towards the product in the market.

2. Review of Literature

A literature review is a comprehensive survey, synthesis, and critical evaluation of existing scholarly research on a specific topic. It maps out current knowledge, identifies key researchers, highlights gaps or contradictions in studies, and provides a theoretical foundation for new research

Key Aspects of a Literature Review:

- Content:** It covers scholarly articles, books, dissertations, and relevant reports.
- Structure:** It is organized by themes, trends, or methodologies rather than just a chronological list of summaries.
- Critical Analysis:** It does not just summarize; it evaluates and compares sources to explain how they relate to the research topic
- Purpose:** To demonstrate familiarity with the field, establish credibility, avoid duplicating previous work, and justify your research question.

Steps to Conduct a Literature Review:

- **Search** for relevant literature.
- **Evaluate** and select relevant sources.
- **Identify** themes, debates, and gaps.
- **Outline** the structure of your review.
- **Write** the review, synthesizing the information

3. Select Appropriate Research Methodology

- Research methods are systematic techniques used to collect and analyze data, generally categorized into quantitative (numerical), qualitative (descriptive), and mixed-methods. Key techniques include surveys for broad data, interviews and focus groups for in-depth insights, observations, and experiments for cause-and-effect relationships.

4. Collecting the data

A data gathering procedure is a systematic, step-by-step plan for collecting accurate, relevant information to answer research questions. It involves defining research objectives, selecting appropriate qualitative or quantitative methods (e.g., surveys, interviews, observations), creating instruments, sampling, gathering data ethically, and organizing data for analysis

5. Interpreting the Data

Data interpretation in research is the process of reviewing, analyzing, and assigning meaning to collected information (qualitative or quantitative) to draw conclusions, identify patterns, and answer research questions. It transforms raw data into actionable insights, bridging the gap between findings and informed, evidence-based decisions or theories.

6. Formulating conclusions.

Based on the analysis the researcher come to a conclusion and either reform his research or arrive at the decision

CHARACTERISTIC OF GOOD RESEARCH



Hypothesis in Research



1. Meaning of Hypothesis

- A **hypothesis** is a tentative statement predicting the relationship between two or more variables.
- Example:
“Higher customer trust increases digital loan adoption.”

Types of Hypothesis

A. Null Hypothesis (H_0): No relationship/effect exists.

Example: Interest rate does not affect demand.

B. Alternative Hypothesis (H_1) Relationship/effect exists.

Example: Interest rate affects demand.

C. Simple Hypothesis: One independent and one dependent variable.

Example: Price affects sales.

D. Complex Hypothesis: Multiple variables involved.

Example: Price, quality, and promotion affect sales.

E. Directional Hypothesis: Specifies direction.

Example: Lower price increases demand.

F. Non-directional Hypothesis: Only states relationship exists.

Example: Price is related to demand.

Steps to Form a Good Hypothesis

1. Identify research problem
2. Review literature
3. Identify variables
4. Establish expected relationship
5. Make clear and testable statement
6. Ensure measurable data availability

Characteristics of Good Hypothesis

1. Clear
2. Specific
3. Testable
4. Logical
5. Based on theory
6. Relevant to objectives

Ethical Issues in Research

What is ethics?

Ethics refers to the philosophical study of moral principles—concepts of right/wrong, good/bad, and virtue—that guide human conduct. It goes beyond law or personal feelings to provide justified, consistent standards (e.g., honesty, fairness, non-maleficence) used to make principled decisions.

Key Aspects of Ethics

Rational Framework: Ethics is not just following rules blindly; it is using reasoning to evaluate *why* an action is right or wrong.

Subject Matter: It focuses on voluntary human actions (choices) that have moral implications, not involuntary behaviors (e.g., sneezing).

Components: Ethics includes core virtues like compassion, loyalty, and integrity





What Are Ethical Issues in Business Research?

Moral Conflicts

Situations where research actions contradict established moral principles or societal values.

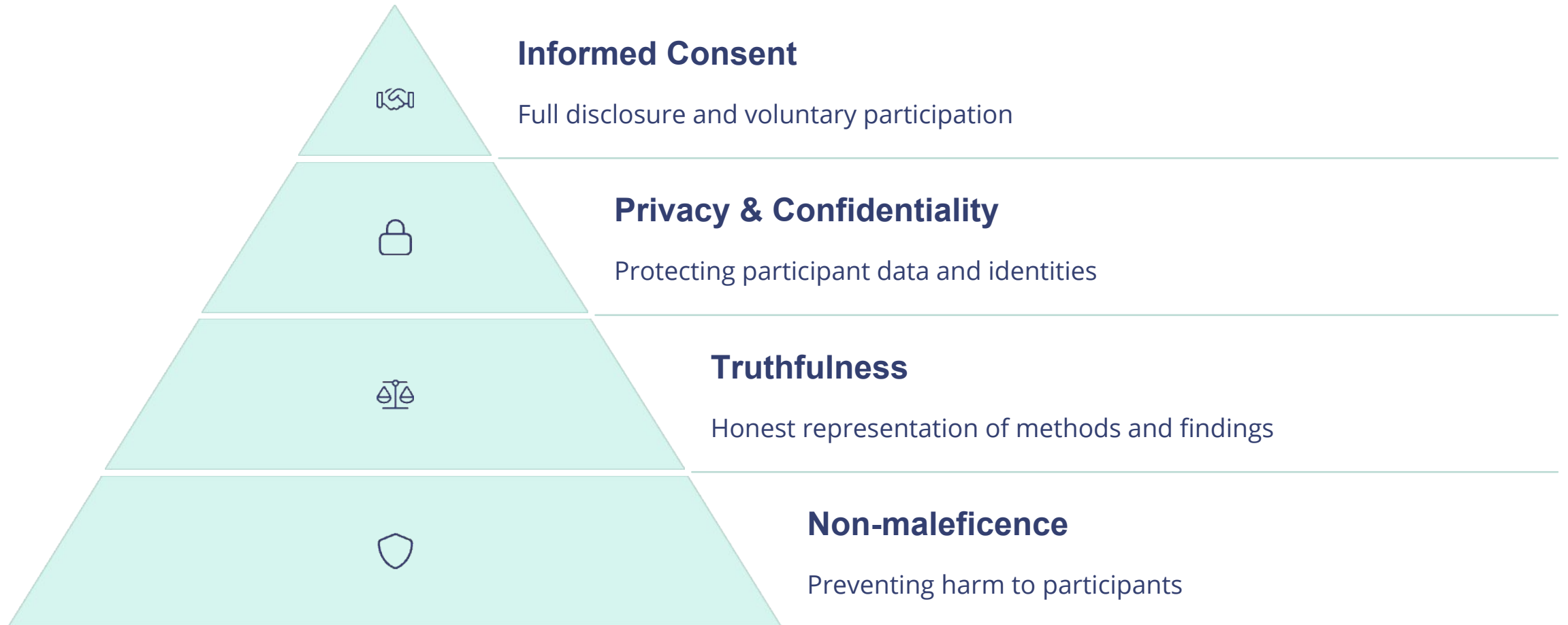
Dual Responsibility

Both individuals and organizations bear ethical obligations in research conduct.

Potential Consequences

Regulatory penalties, damaged reputation, and stakeholder backlash from ethical breaches.

Key Ethical Principles in Business Research



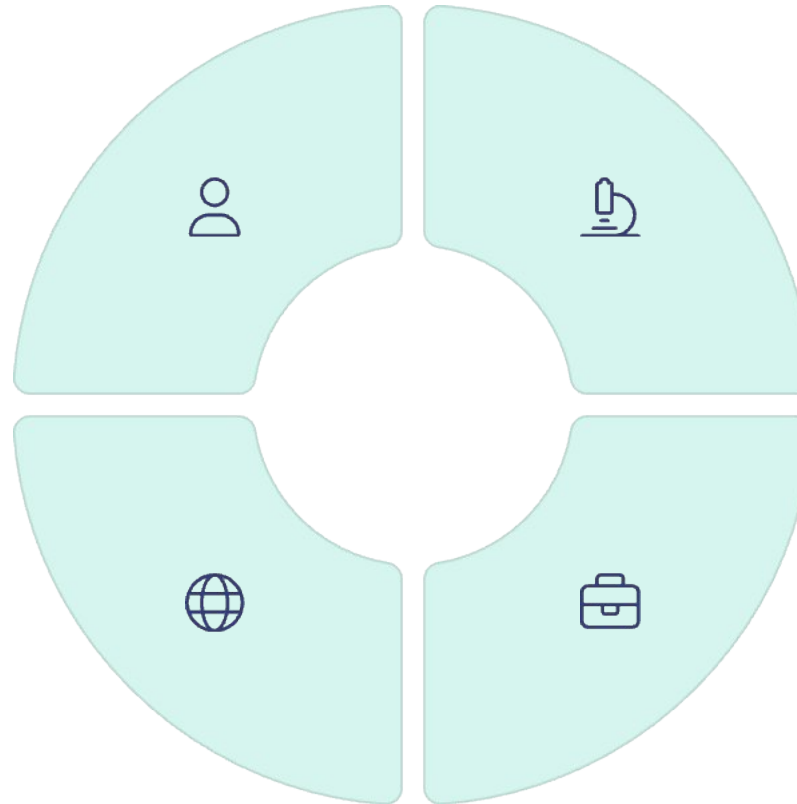
Stakeholders and Their Rights

Respondents

- Data privacy protection
- Accurate study information
- Right to withdraw

Society

- Beneficial outcomes
- Truth in reporting
- Responsible innovation



Researchers

- Professional standards
- Methodological integrity
- Objective analysis

Sponsors/Clients

- Unbiased findings
- Resource accountability
- Contractual transparency



Common Unethical Practices



Confidentiality Breaches

Sharing identifiable participant data without permission or proper anonymization.



Data Falsification

Inventing, altering, or selectively reporting research findings to support desired outcomes.



Misrepresentation

Distorting research conclusions to satisfy corporate agendas or personal interests.



Procedural Violations

Bypassing required ethical reviews or conducting research without proper approvals.

Real-World Examples of Ethical Dilemmas



Customer Data Misuse

Market research firm repurposed confidential customer survey responses for secondary analysis without consent.



Innovation. Questioned.

Volkswagen Emissions Scandal

Researchers deliberately falsified emissions test data to show compliance with environmental regulations.



Privacy Violations

Consumer research company failed to anonymize sensitive participant data before sharing with third parties.

Managing and Preventing Ethical Issues



Ethics Policies

Develop clear organizational codes and guidelines for research conduct.



Training

Educate researchers on ethical principles and real-world application.



Monitoring

Implement review boards and regular compliance audits.



Reporting Systems

Create safe channels for reporting ethical concerns without fear.





**New ethical beginnings
in research.**

Conclusion: The Path Forward

Reinforce Ethics as Core Value

Prioritize integrity over expedience. Ethical research builds lasting credibility and trust.

Commit Organizationally

Dedicate resources to ethical oversight. Make ethical standards part of performance evaluation.

Foster Ethical Culture

Celebrate ethical choices. Create environments where integrity is rewarded over shortcuts.

THANK YOU

Module-2

Business Research Design

Presentation by
Dr. Basavaraj S.Kudachimath

Discussion Points.

- **Business Research Design:** Meaning, types and significance of research design, errors affecting research design.
- **Exploratory Research:** Meaning, purpose, methods, Literature search, experience survey, focus
- groups and comprehensive case methods.
- **Conclusive Research Design:** Descriptive Research, Meaning, Types, Cross sectional studies and longitudinal studies.
- **Experimental Research Design:** Meaning and classification of experimental designs, formal and informal, Pre experimental design, True experimental design, Quasi-experimental design, Statistical experimental design.

What is Research design?

Research design could be defined as the blue print specifying every stage of action in the course of research. Such a design would indicate whether the course of action planned will minimize the use of resources and maximize the outcome. Research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure.

Research design would answer the following questions :

- What is the study about ?
- Why is the study being made?
- Where will the study be carried out?
- What type of data and where it would be collected ?
- What is the period of study ?
- Whether any sample would be used and if so, what type of sample will be used ?
- What type of analytical tools will be used ?

Parts of Research Design:

1. Objectives and hypothesis design
2. Sampling design – all the details connected with the sampling process, from the determination of sample size down to the collection of data, would be spelt out.
3. Observational design – if the study makes use of observational technique, then what type of observation technique would be used, conditions under which the observation will be made, would be indicated.
4. Statistical design – this part of the research design would spell out the type of analysis that would be carried out.
5. Operational design – this design would lay down the steps that would be taken at each stage as the design is executed.

Criteria of Good Research Design

1. It is flexible. As and when it is called for the research design should be amenable for accommodating any change.
2. It should help to minimize bias at every stage.
3. It should facilitate collection and analysis of maximum information.
4. The design should be closely linked with the objectives of the study.

Types of Research Design

1.Exploratory 2. Descriptive 3. Experimental

1. Exploratory Research Design (Formulative Research Design) : A design for exploratory research work is called Exploratory Research Design. Exploratory Research is done to discover new insights or ideas in the existing body of knowledge. It is conducted when available information in literature is not sufficient to address the problem.

Exploratory research answers "what," "how," and "why" questions regarding poorly understood or new topics

Ex.1. Juice owner wanted to know the demand for introducing new juice. He does not know the consumer demand.

2. A resaerch on terror attack in Phulvama.

Types of exploratory research design

1. Primary Research Methods of collecting data: (More of qualitative in nature)
 - a. Projective Technique (Individual are asked to express opinion or feelings)
 - b. Depth Interview
 - c. Experience survey
 - d. Focus group
 - e. Observation techniques
 2. Secondary Research Method of Collecting data
 - a. Survey of related literature -
 - b. Conducting experience survey - this refers to undertaking collection of details and discussion with the experienced people in the chosen field of research. This would help the researcher to determine the extent to which he is original and can avoid duplication.
- Analysis of insight-stimulating examples is yet another method, in which depending upon the study on hand. In this method, the experience of people would be used as a guide to develop or formulate a hypothesis.

Uses of Exploratory Research design

1. Most useful in investigating for new things.
2. Gaining information in detail.
3. Clarification of doubts.
4. Analysing the market trends.
5. Exploring new approaches

2. Descriptive Research Design (Diagnostic/ Conclusive)

Descriptive research design is concerned with the research studies with a focus on the portrayal of the characteristics of a group or individual or a situation. The main objective of such studies is to acquire knowledge.

For example, to identify the use of a product to various groups, a research study may be undertaken to question whether the use varies with income, age, gender or any other characteristics of population. Similarly such studies are used to examine the characteristics of the corporate sector or consumer behavior etc.,

This research helps to answer the questions What? How? When? Where? But does not answer Why?

Types of Descriptive Research Design

1. **Cross Sectional Research:** Data is collected from population at a specific point of time from different samples.
 - **Examples:**
 - A survey conducted in 2025 to compare **stress levels of college students in different cities**.
 - Studying **customer satisfaction** of various brands by collecting responses from different customers at the same time.
 - Measuring **health conditions (e.g., obesity rates)** across different age groups in a population during one specific year.
 - A company analyzing **employee job satisfaction across departments** at one point in time.
2. **Longitudinal Research:** When the data is collected from the same sample population different data for different time interval.
 - **Examples:**
 - Tracking the **same group of students' academic performance** from 1st year to final year.
 - Studying **weight loss progress** of individuals over 6 months after starting a diet program.
 - Observing **customer loyalty behavior** of the same customers over several years.
 - Monitoring **impact of a new drug** on the same patients at monthly intervals.

Uses of Descriptive Research

1. Highlight the social problem
2. Describe the real life situation.
3. Helps to gain knowledge about existing phenomena.

Exploratory vs Descriptive Research

Basis for Comparison	Exploratory Research	Descriptive Research
Meaning	Exploratory research means a research conducted for formulating a problem for more clear investigation. It is generally conducted when a little is known about a topic of interest.	Descriptive research is a research that explore and explain an individual, group or a situation.
Objective	To provide insights and understanding and develop hypothesis	To test specific hypothesis, examine relationships, and describe characteristics and functions
Overall Design	Flexible	Rigid
Research Process	Unstructured	Structured
Sampling	Non-probability sampling	Probability sampling
Structural Design	No pre-planned design for analysis	Pre-planned design for analysis
Findings/Results	Tentative	Conclusive
Outcome	Generally followed by further exploratory or conclusive research	Findings can be used as input for decision making

3. Experimental Research Design

Research method in which conditions are controlled so that one or more independent variables can be manipulated to test the hypothesis about dependent variables.

Allows evaluation of causal relationships among variables while other variables are eliminated or controlled.

Example: If you provide discount on the price what is its effect on sales.

Defining Experimental Research

Experimental research design is concerned with examination of the effect of independent variable on the dependent variable, where the independent variable is manipulated through treatment and the effect of those interventions is observed on the dependent variable.

The experimental research studies are mainly focused on finding out the cause and effect relationship of the phenomenon under study.

Actually, when observation is arranged and controlled, it becomes experimental study.

An experiment is a test or trial or an act or operation for the purpose of discovering something unknown or of testing a principle, supposition, etc. It is a process in which one or more variables are manipulated under conditions that permit the collection of data that show the effects of any of such variables in a unconfused fashion.

There are different types of experiments, such as : methodological, pilot study, heuristic, fact finding, boundary, simulation, theoretical, illustrative, etc.

Concepts relating to Experimental Research Design

1. Dependent and Independent variable.
2. Extraneous Variable: Independent variables that are not related to the purpose of the study but may affect the dependent variable. It leads to error in study.
3. Control: Minimize the influence of Extraneous variable on the study.
4. Confounded relationship: When the dependent variable is not free from Extraneous variable.

Types of Experimental Design:

Types of experimental Research Designs

1. Pre-Experimental Research Design
2. True- Experimental Research Design
3. Quasi Experimental Research Design

Types of Experimental Research Designs

- 1. Pre-experimental Research Design:** A research study could conduct pre-experimental research design when a group or many groups are under observation after implementing factors of cause and effect of the research. The pre-experimental design will help researchers understand whether further investigation is necessary for the groups under observation.

Example:

A teacher introduces a new teaching method to one class and observes students' performance after one month.

- 2. True Experimental Research Design:** A true experimental research design relies on statistical analysis to prove or disprove a researcher's hypothesis. It is one of the most accurate forms of research because it provides specific scientific evidence. Furthermore, out of all the types of experimental designs, only a true experimental design can establish a cause-effect relationship within a group.

Example:

A pharmaceutical company randomly divides patients into two groups: one receives a new medicine, while the other receives a placebo to measure the medicine's effectiveness.

3. Quasi-experimental Research Design: The word “Quasi” means similarity. A quasi-experimental design is similar to a true experimental design. However, the difference between the two is the assignment of the control group. In this research design, an independent variable is manipulated, but the participants of a group are not randomly assigned. This type of research design is used in field settings where random assignment is either irrelevant or not required.

Example:

A school compares the academic performance of two existing classes where one class uses smart boards and the other follows traditional teaching methods.

Advantages of ERD

1. Researchers have firm control over variables to obtain results.
2. The subject does not impact the effectiveness of experimental research. Anyone can implement it for research purposes.
3. The results are specific.
4. Post results analysis, research findings from the same dataset can be repurposed for similar research ideas.
5. Researchers can identify the cause and effect of the hypothesis and further analyze this relationship to determine in-depth ideas.
6. Experimental research makes an ideal starting point. The collected data could be used as a foundation to build new research ideas for further studies.

Class preparation:

Which of the research design is applicable for the following topics. Explain with reasons and also draft a research design for each of the following.

1. Exploring the Impact of Artificial Intelligence Tools on College Students' Learning Habits.
2. A Study on Customer Satisfaction Towards Online Food Delivery Services.
3. Impact of Artificial Intelligence (AI)-Based Recruitment Tools on the Hiring Efficiency of Organizations.

Errors in Research Design

A useful way of looking at the research process is that involves the management of errors. At all stages, from problem formulation to report presentation errors can arise. It is rare that a research project will be error free. Consequently the research designer must adopt a strategy for managing and maintaining the error. We shall first look at the component of errors and then the type of errors.

Components of Errors in Research

1. Sampling Error: refers to the variable error resulting from chance selection of elements from population as per the sampling plan. Since it introduces random variability into the precision with which sample statistics are calculated it is also called random sampling error.
2. Non Sampling error.: consists of all other error associated with the research project. Such errors are diverse in nature. They are often thought of as bias. However bias is a type of systematic error which enters into the process because of un-calibrated instruments or prejudices of the researcher.

Types of Errors in research

A. Sampling error

1. Population specification error.
2. Sampling Selection error
3. Sampling frame error
4. Non Response error.
5. Surrogate Information Error.

B . Non Sampling Error.

1. Measurement error
2. Experimental Error

• Notes below

Thank You

UNIT-3

SAMPLING TECHNIQUES



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Discussion Points

- Sampling: Concepts,
- Types of Sampling,
- **Probability Sampling:** simple random sampling,
- systematic sampling, stratified random sampling, cluster sampling,
- **Non Probability Sampling:** convenience sampling- judgmental sampling, snowball sampling, quota
- sampling, Errors in sampling.

What is Sampling?

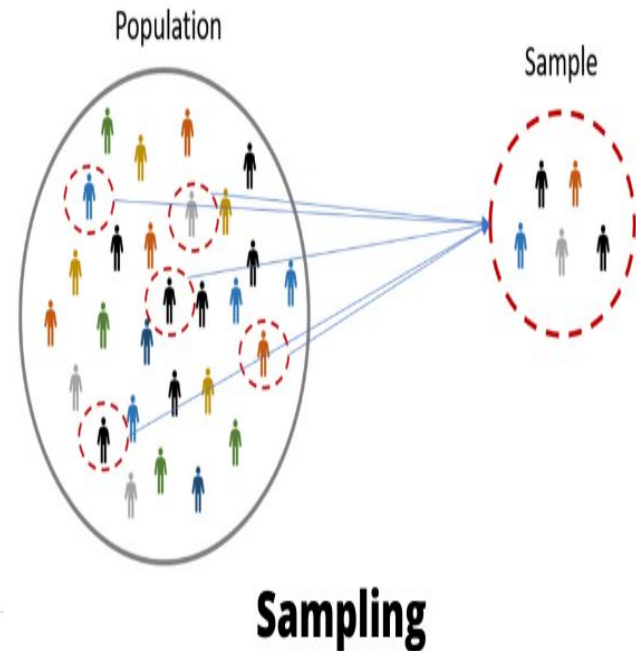
Procedure by which some members of a given population are selected as representatives of the entire population.



PURPOSE OF SAMPLING

To gather data about the population in order to make an inference that can be generalized to the population.

- **POPULATION**
- **INFERENCE**
- **SAMPLE**



Process Of Sampling

1. Define the Population
2. Develop Sampling Frame
3. Select a Sampling Method
4. Determine Sample Size
5. Execute the Sampling Process



Who do you want to generalize to?

The Theoretical Population

What population can you get access to?

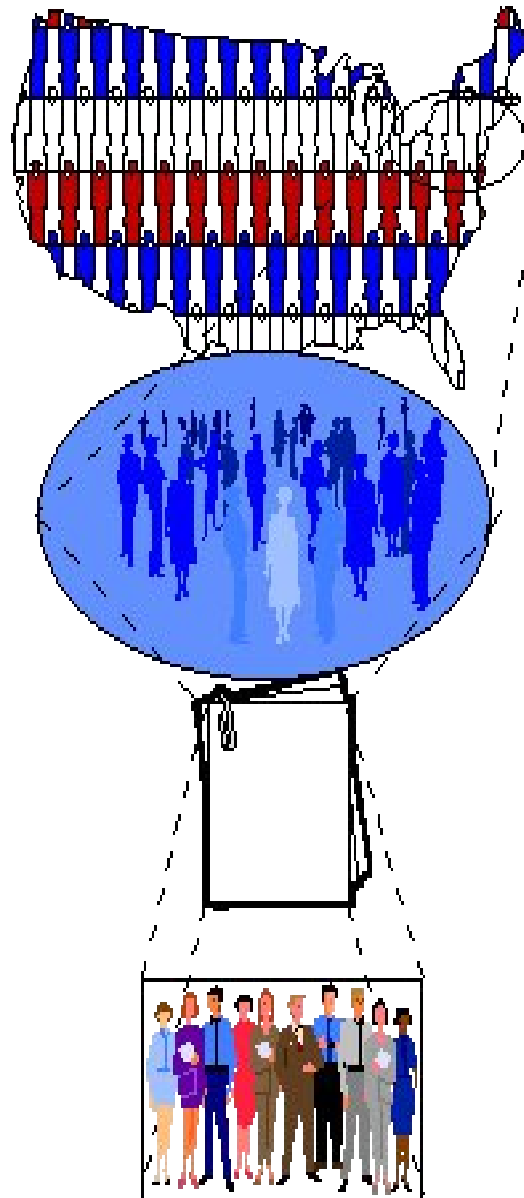
The Study Population

How can you get access to them?

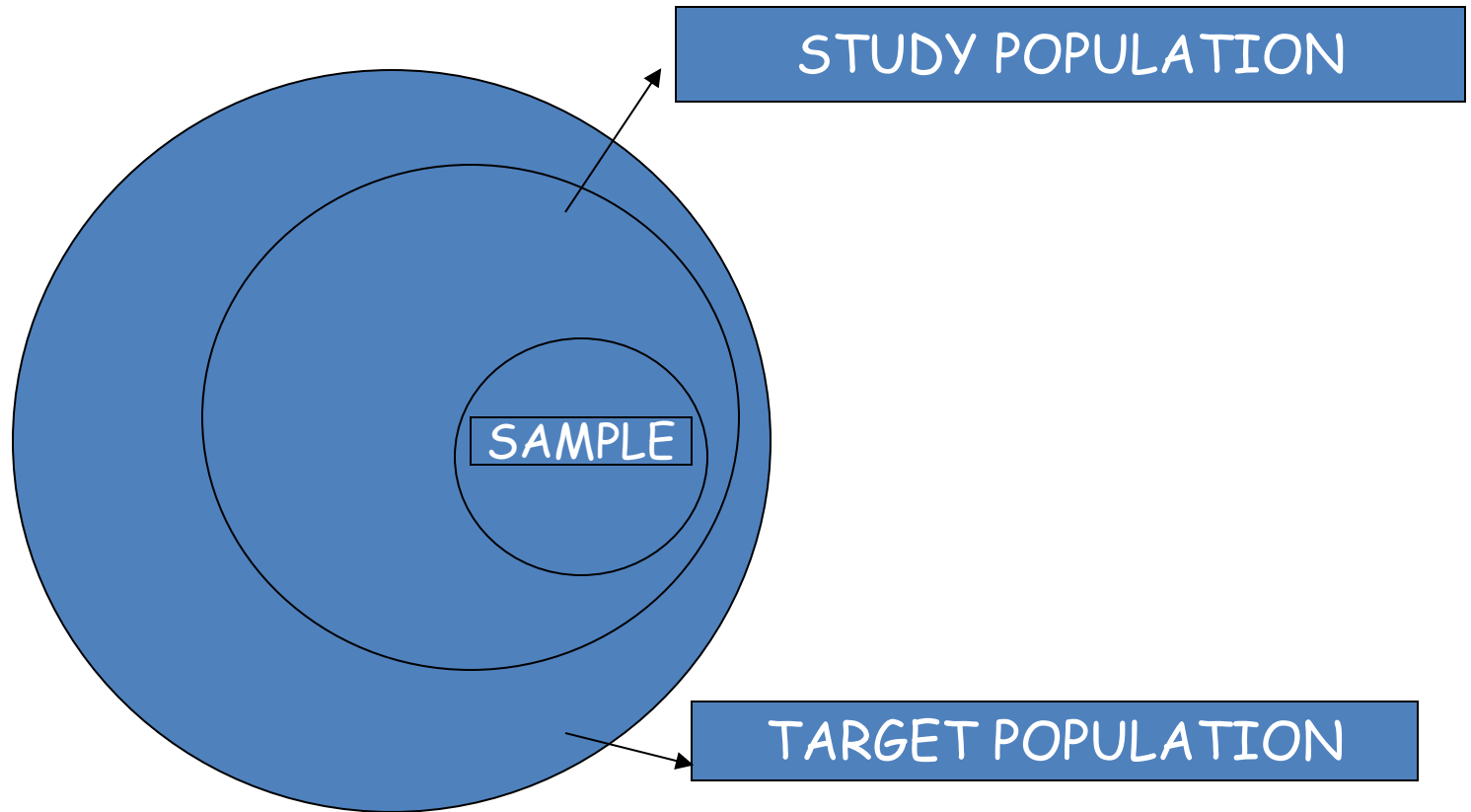
The Sampling Frame

Who is in your study?

The Sample



SAMPLING.....



Types of Samples

I. Probability Samples

Simple random sample

Systematic random sample

Stratified random sample

Cluster sample

Multistage sample

II. Non-Probability Samples

– Convenience sample

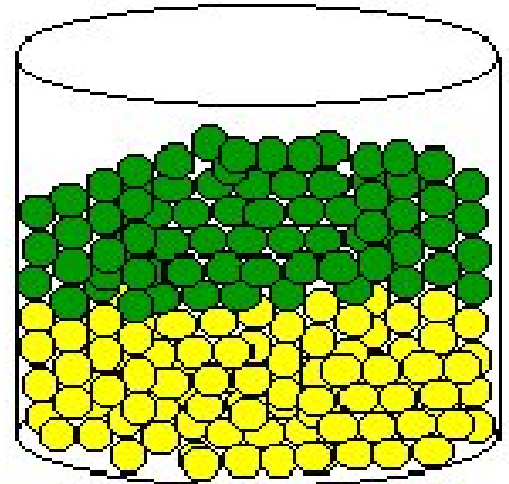
– Purposive sample

– Quota

– Snow ball sampling

PROBABILITY SAMPLING

A **probability sampling** scheme is one in which every unit in the population has a chance (greater than zero) of being selected in the sample, and this probability can be accurately determined.



PROBABILITY SAMPLING.....

Types:

- Simple Random Sampling,
- Systematic Sampling,
- Stratified Random Sampling,
- Cluster Sampling
- Multistage Sampling.

NON PROBABILITY SAMPLING

- Any sampling method where some elements of population have *no* chance of selection (these are sometimes referred to as 'out of coverage'/'undercovered'), or where the probability of selection can't be accurately determined. It involves the selection of elements based on assumptions regarding the population of interest, which forms the criteria for selection. Hence, because the selection of elements is nonrandom, nonprobability sampling not allows the estimation of sampling errors..



NONPROBABILITY SAMPLING

Nonprobability Sampling includes:

- Convenience Sampling,
- Quota Sampling
- Judgemental Purposive Sampling.
- Snow ball Samling

TYPES OF PROBABILITY SAMPLING.....

1. Simple Random Sampling,
2. Systematic Sampling,
3. Stratified Random Sampling,
4. Cluster Sampling
5. Multistage Sampling.

1. SIMPLE RANDOM SAMPLING

- Applicable when population is small, homogeneous & readily available
- All subsets of the frame are given an equal probability. Each element of the frame thus has an equal probability of selection.
- It provides for greatest number of possible samples. This is done by assigning a number to each unit in the sampling frame.
- A table of random number or lottery system is used to determine which units are to be selected.



2. SYSTEMATIC SAMPLING

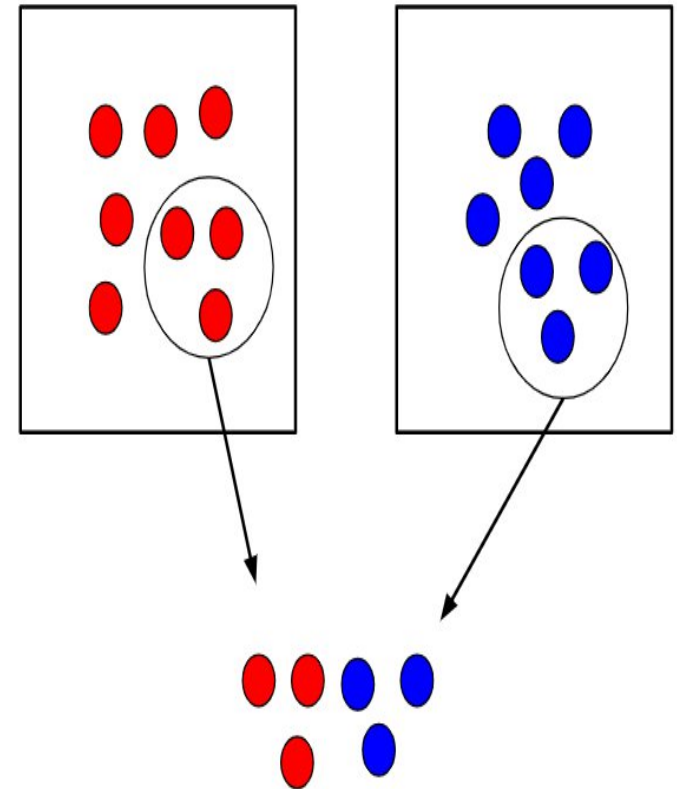
- Systematic sampling relies on arranging the target population according to some ordering scheme and then selecting elements at regular intervals through that ordered list.



3. STRATIFIED SAMPLING

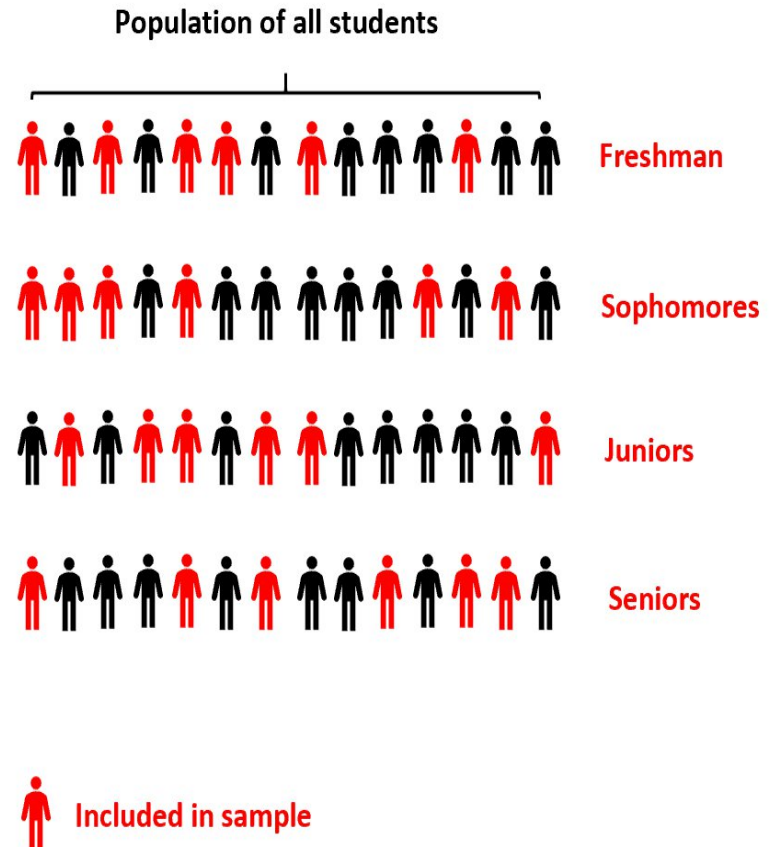
Where population embraces a number of distinct categories, the frame can be organized into separate "strata." Each stratum is then sampled as an independent sub-population, out of which individual elements can be randomly selected.

- Every unit in a stratum has same chance of being selected.
- Using same sampling fraction for all strata ensures proportionate representation in the sample.
- Adequate representation of minority subgroups of interest can be ensured by stratification & varying sampling fraction between strata as required.



4. CLUSTER SAMPLING

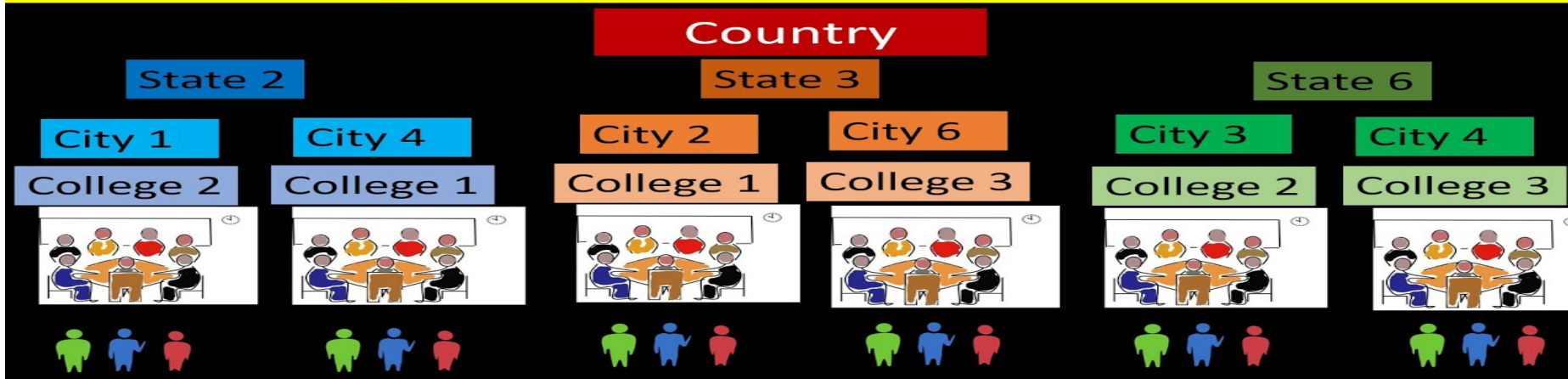
- First stage a sample of areas is chosen;
- Second stage a sample of respondents *within* those areas is selected.
- Population divided into clusters of homogeneous units, usually based on geographical contiguity.
- Sampling units are groups rather than individuals.
- A sample of such clusters is then selected.
- All units from the selected clusters are studied.



5. MULTISTAGE SAMPLING

- Complex form of cluster sampling in which two or more levels of units are embedded one in the other.
- First stage, random number of districts chosen in all states.
- Followed by random number of talukas, villages.
- Then third stage units will be houses.

MULTI-STAGE SAMPLING



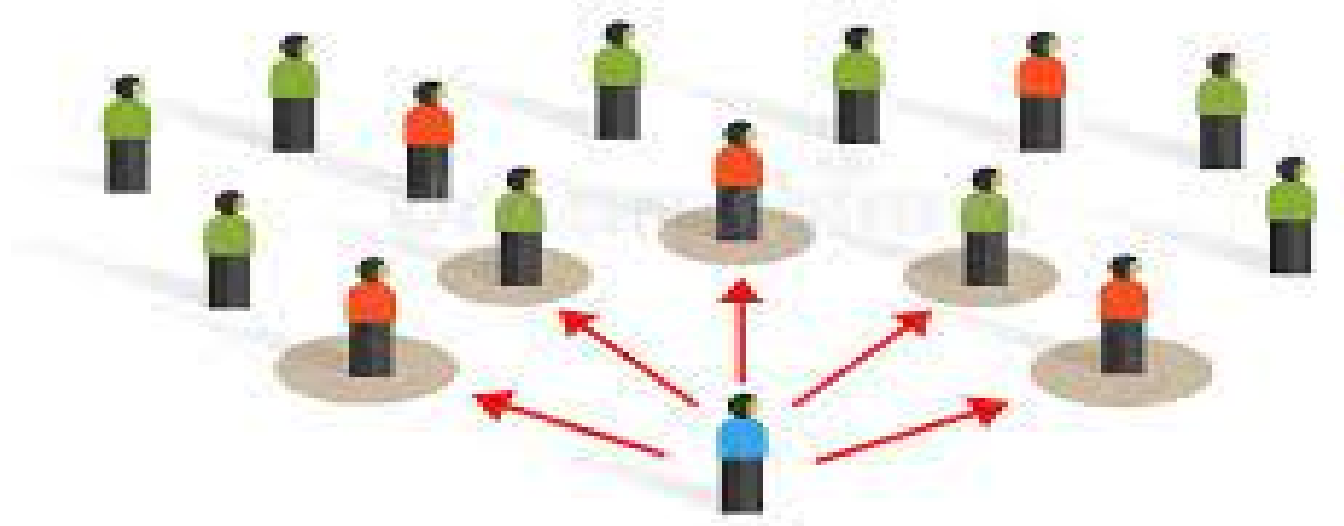
TYPES OF NONPROBABILITY SAMPLING

1. Convenience Sampling,
2. Quota Sampling
3. Judgemental /Purposive Sampling.
4. Snow ball Samling

5. CONVENIENCE SAMPLING

- Sometimes known as **grab** or **opportunity sampling** or **accidental** or **haphazard sampling**.
- A type of nonprobability sampling which involves the sample being drawn from that part of the population which is close to hand. That is, readily available and convenient.
- The researcher using such a sample cannot scientifically make generalizations about the total population from this sample because it would not be representative enough.

Convenience sampling



2. QUOTA SAMPLING

- The population is first segmented into mutually exclusive sub-groups, just as in stratified sampling.
- Then judgment used to select subjects or units from each segment based on a specified proportion.



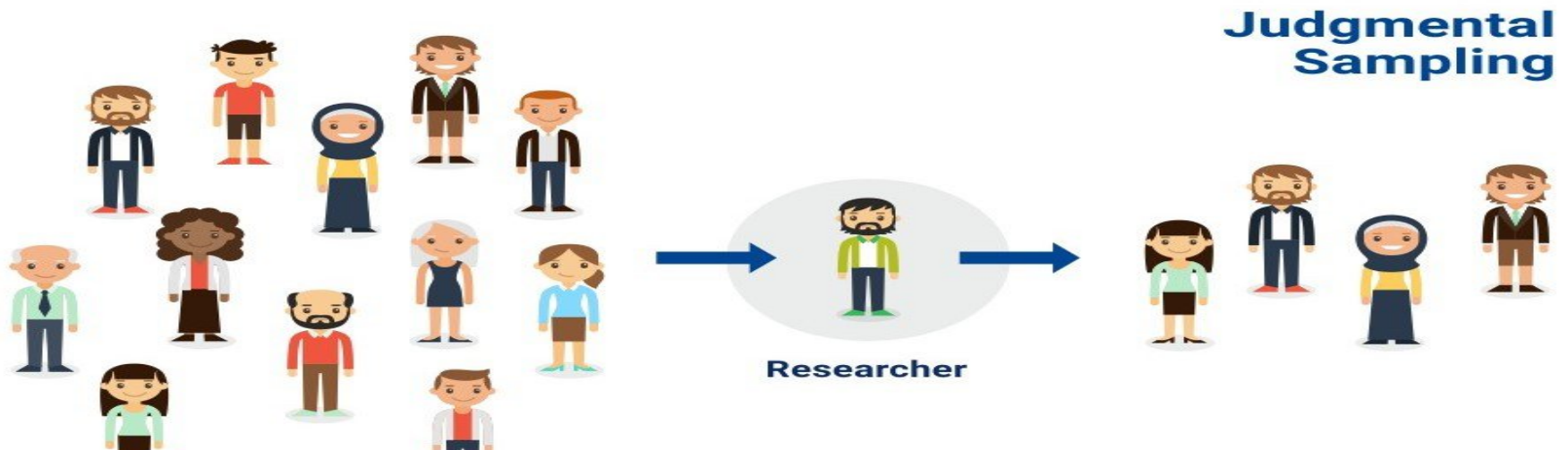
Symbol	Age Group	No.
	11-21 Years	11
	22-31 Years	16
	32-41 Years	15
	42-51 Years	18
Total	11-51 Years	60

Quota Sampling

3. Judgmental sampling or Purposive sampling

The researcher chooses the sample based on who they think would be appropriate for the study.

This is used primarily when there is a limited number of people that have expertise in the area being researched



4. Snowball sampling

Snowball sampling is finding the large samples with the help of small sample. It is used where potential participants are hard to find. It's called snowball sampling because (in theory) once you have the ball rolling, it picks up more “snow” along the way and becomes larger and larger.



Questions???



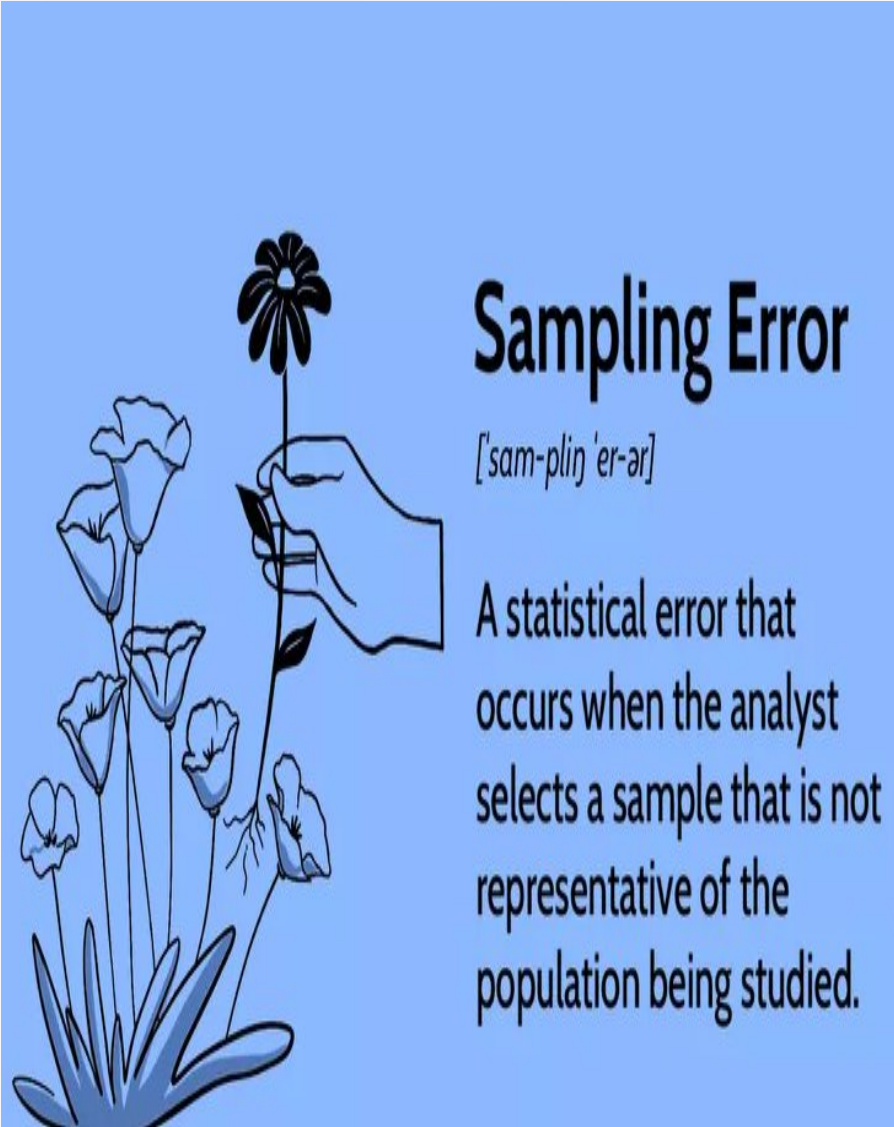
What sampling method do you recommend?

- A study on undernourished five year old children in Jamboti Village in Belgaum district.
- Investigating nutritional status of preschool children in Karnataka state.
- Selecting maternity records for the study of previous abortions or duration of postnatal stay.
- A study on impact of Covaccine protection against Covid-19 disease
- Study of consumer behavior towards Big Bazaar in Belgaum.



Understanding Sampling Errors

Bridging the gap between **sample** and **population** — why the data we collect rarely tells the whole story.



Sampling Error

['sam-plɪŋ 'er-ər]

A statistical error that occurs when the analyst selects a sample that is not representative of the population being studied.

What is a Sampling Error?

A sampling error is the **gap between what your sample tells you and what is actually true** about the entire population.

It is the difference between a **sample statistic** (e.g., sample mean) and the true **population parameter** (e.g., population mean).

These errors are inherent in any study that uses a sample instead of a complete census — making them impossible to eliminate entirely, but possible to minimise.

Types of Sampling Errors

Selection Bias

When the sampling method is not random, the sample **systematically differs** from the population.

Example: Surveying only smartphone users to understand general internet usage — excluding those without smartphones.

Sample Size Too Small

A sample that is too small may **fail to capture the diversity** of the population.

Example: Estimating average city height by measuring only 10 people.

Disproportionate Representation

The sample's proportions of certain characteristics **do not match** the population's proportions.

Example: A consumer survey skewed heavily towards young adults when they are a minority in the real population.

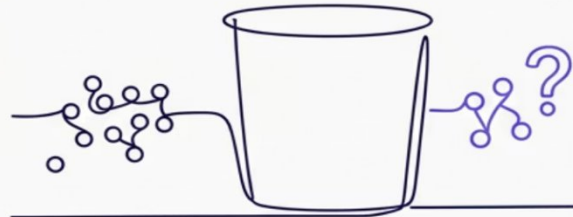


Causes of Sampling Errors



1. Non-Random Sampling Methods

Convenience or voluntary response sampling introduces bias.



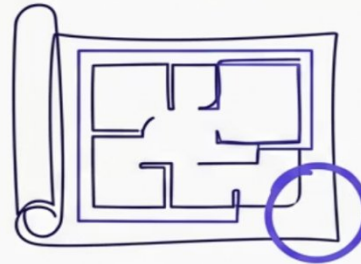
2. Inadequate Sample Size

Too few data points to capture population variability.



3. Population Variability

High diversity makes representation harder.



4. Flawed Sample Design

Plan does not account for key population characteristics.

Real-World Examples of Sampling Errors



Political Polling

A pre-election poll that **over-samples a particular demographic** can inaccurately predict the election outcome — as seen in several high-profile electoral upsets.



Market Research

Surveying only **existing customers** to gauge interest in a new product misses the perspectives of potential new customer segments entirely.



Health Studies

A drug efficacy trial limited to one **geographic region** may not reflect how the treatment performs across diverse populations with different genetic or environmental profiles.



Minimising Sampling Errors

While sampling errors can never be fully eliminated, following these principles drives your research **towards greater accuracy and reliability.**

01

Increase Sample Size

Larger samples reduce sampling error and better reflect population variability.

02

Use Random Sampling Techniques

Apply simple random, stratified, or cluster sampling to minimise bias.

03

Ensure Proportional Representation

Mirror the population's key demographic and characteristic proportions in your sample design.

04

Define the Target Population Clearly

Precisely identify who you are studying before selecting a single respondent.



Unit-4

Data Collection

Presentation by
Dr. Basavaraj S. Kudachimath

Discussion Points

- **Data Collection:** Meaning, types, **Data collection methods:** Observations, survey and interview techniques,
- **Questionnaire design:** Meaning, process of designing questionnaire. Qualitative Techniques of data collection Secondary data Sources: advantages and disadvantages.
- **Measurement and Scaling Techniques:** Basic measurement scales- Nominal scale, Ordinal scale, Interval scale, Ratio scale. Attitude measurement scale - Likert Scale, Semantic Differential Scale, Thurston scale,
- **Multi-Dimensional Scaling:** Non comparative scaling techniques

What is Data

- Data is defined as a collection of individual facts or statistics. (While “datum” is technically the singular form of “data,” it’s not commonly used in everyday language.)
- Data are **facts such as figures or numbers, pictures, signs, colours etc used to analyze something or make decisions.**
- Data can come in the form of **text, observations, figures, images, numbers, graphs, or symbols.** For example, data might include individual prices, weights, addresses, ages, names, temperatures, dates, or distances. Data is a raw form of knowledge and, on its own, doesn't carry any significance or purpose.

There are two main types of data:

1. **Quantitative data** is provided in numerical form, like the weight, volume, or cost of an item.
2. **Qualitative data** is descriptive, but non-numerical, like the name, sex, or eye color of a person.

Information

- **What Is Information?**
- Information is defined as knowledge gained through study, communication, research, or instruction. Essentially, information is the result of analyzing and interpreting pieces of data. Whereas data is the individual figures, numbers, or graphs, information is the perception of those pieces of knowledge.
- For example, a set of data could include temperature readings in a location over several years. Without any additional context, those temperatures have no meaning. However, when you analyze and organize that information, you could determine seasonal temperature patterns or even broader climate trends. Only when the data is organized and compiled in a useful way can it provide information that is beneficial to others.

Differences between Data and Information

Data

- A collection of facts or statistics
- Unorganized
- Without context
- Can be quantitative (numerical) or qualitative (descriptive)
- Examples: number of website visitors, individual customer survey responses, product price

Information

- The result of analyzing and interpreting data
- Organized
- Has context
- Can be used to help make decisions
- Examples: website traffic changes, customer sentiment based on survey results, product price comparison

Types of data

1. Primary Data: It may be

- i) qualitative data or
- ii) quantitative data

1. Secondary data: It may be

- i) qualitative data or
- ii) quantitative data

Primary Data

Primary data is known as the data collected for the first time through field survey. Such data are collected with specific set of objectives to assess the current status of any variable studied.

Primary data always reveal the cross section picture of any thing studied. This is needed in research to study the effect or impact of any policy.

For example, suppose the Transport department wants to study whether introduction of one way regulation in a main road has helped to release the traffic congestion. This can be studied only by collecting information on the spot for a few days from the common public, vehicle using population, shop keepers, policemen on duty and others who are directly affected by such a one way regulation. Such an information is called primary data.

Advantages of Primary data

1. Gives more accuracy of research.
2. Gives live input for decision making.
3. Easy to settle queries.
4. Latest data makes research more reliable.

Disadvantages

1. Not suitable for all types of research.
2. Difficult to check accuracy of research.
3. Researcher should be efficient enough to collect the data

Sources of Primary Data

1. Schedule or Questionnaire (Survey method)
2. Interview
3. Observation
4. Feedback Form
5. Sales force Opinion
6. Pantry Audit.
7. Consumer Panel
8. Collection through mechanical device.

Survey method tools

Schedule & questionnaire

The survey method is a primary data collection technique that gathers information from a targeted group of respondents using structured schedules or questionnaire.

Meaning & Types of schedule

Schedule is a sheet of paper containing questions relating to certain specific aspect, about which the scholar is collecting data. But the schedule is not sent to the informant. It is taken by the researcher, used by him while eliciting information. The reply is recorded by the researcher himself against each question. Hence, the schedules are different from questionnaires only in respect of usage.

Types of Schedules

1. Interview Schedule

Used in face-to-face or remote interactions, where the enumerator asks the questions exactly as written and records the respondent's answers.

Structured: Features predetermined questions in a strict order.

Unstructured / In-depth: Provides a framework of topics to guide a conversation, allowing for flexible follow-up questions.

3. Rating Schedule

Designed to measure the trend, value, or intensity of qualitative elements such as attitudes, opinions, preferences, or inhibitions.

These are heavily utilized in psychology and business to quantify subjective traits (e.g., measuring satisfaction levels using numerical scales or scoring).

5. Institutional or Evaluation Schedule

Used to gather in-depth data about specialized organizations, agencies, or institutions.

The size and complexity of this schedule scale with the nature and complexity of the institution being evaluated

2. Observation Schedule

Utilized when the researcher's primary method is observing behaviors, activities, or conditions rather than directly asking questions.

The observer watches a subject (e.g., in a classroom, clinic, or community) and records specific actions on a standardized form

4. Document Schedule

Used to extract specific information from historical records, written evidence, archives, and case histories. Rather than asking people, the enumerator gathers data by tracking how often specific terms or functional issues occur in existing documents.

6. Demographic/Household Schedule

Commonly used in large-scale social surveys and census research to collect general information about a community or family unit.

Gathers full demographic details including age, sex, occupation, education, and family relations.

Dr Bhatras schedule for investigating alopecia in patients

• To be filled by primary investigator (Enumerator). Name: _____ Date_____

• Proforma for Investigating Alopecia

1. Patient History

A. Onset and duration of hair loss (acute, chronic, sudden, gradual)

A. Ans _____

B. Pattern of hair loss (diffuse, patchy, focal, patterned)

A. Ans _____

C. Associated symptoms: itching, pain, burning, scaling, erythema

A. Ans _____

D. Personal and family history of hair disorders or autoimmune diseases

A. Ans _____

E. Recent illnesses, stress, or major life events.

A. Ans _____

F. Medications (including recent changes)

A. Ans _____

- **2. Physical Examination (Densitometry test)**

- Densitometry scalp and hair inspection: distribution, pattern, and extent of hair loss; presence of scarring, scaling, erythema, pustules, or atrophy.

Report result _____

- Examination of other hair-bearing areas: eyebrows, eyelashes, beard, body hair

Ans _____

- Nail examination: pitting, ridging, other abnormalities

Ans _____

- Skin examination: look for signs of systemic disease (e.g., lupus, lichen planus) etc

Ans _____

3. Diagnostic Tests

- **Hair pull test:** Gently pull 20–60 hairs; >10% release suggests active shedding
 - Ans _____
- **Hair tug test:** Assesses hair shaft fragility
 - Ans _____
- **Trichoscopy (dermatoscopy):** Non-invasive visualization of hair and scalp for characteristic features (e.g., yellow dots, exclamation mark hairs, follicular ostia) **Card test:** Assesses new hair growth and shaft abnormalities.
 - Ans _____

4. Laboratory Investigations

- **Complete blood count (CBC)**
 - Test Results _____
- **Thyroid function tests (TSH, free T4)**
 - Test Result _____
- **Serum ferritin (iron stores)**
 - Ans _____
- **Vitamin D levels**
- Ans _____

5. Additional Diagnostic Procedures

- **Scalp biopsy:** If diagnosis is unclear or scarring alopecia is suspected (usually a 4-mm punch biopsy from the active margin)

Ans _____

- **Fungal studies:** KOH preparation and/or fungal culture if tinea capitis is suspected

Ans _____

- **Microscopy of hair roots:** To distinguish between anagen and telogen hairs, or identify shaft abnormalities.

Test results _____

Advantages of Schedules

1. **Suitable for the illiterate:** Because an enumerator reads the questions and records the answers, this method can easily gather data from respondents who cannot read or write.
2. **High response rates:** Direct, face-to-face interaction virtually eliminates the problem of non-response since the enumerator is present to prompt answers to all questions.
3. **Fewer misunderstandings:** If a respondent misinterprets a question, the enumerator can clarify the context immediately, ensuring more accurate data.
4. **Complete information:** Enumerators ensure that no questions are skipped and the gathered data is comprehensive.

Disadvantages of Schedules

1. **Costly:** Hiring, training, and deploying a team of enumerators to the field requires significant financial investment.
2. **Time-consuming:** Collecting data face-to-face over large geographic areas or sample sizes is considerably slower than sending out mass mail questionnaires.
3. **Interviewer bias:** The presence and demeanor of the enumerator, or their potential misinterpretation of responses, can consciously or unconsciously influence the data.
4. **Requires highly trained staff:** The reliability of the data depends entirely on the honesty, dedication, and competence of the field enumerators

Questionnaire

A questionnaire is a sheet [s] of paper containing questions relating to certain specific aspect, regarding which the researcher collects the data. The questionnaire is given to the informant or respondent to be filled up. Sometimes questionnaire is also in the form of files generated through computer. This is usually called soft copy of questionnaire.

Advantages of Questionnaire Method

1. It involves lesser cost as questionnaire could be sent by post to a wide area
2. It does not interfere with the respondent while answering the question
3. The bias of the investigator is completely eliminated.
4. Respondents are given sufficient time to fill up the questionnaire.
5. If respondents cannot be reached personally, questionnaire is the alternative method available.
6. This method is useful when the sample size is very large
7. If the questionnaire is designed properly with instructions, then the time and cost of recruiting, training, supervising and controlling the investigators is saved.
8. Confidentiality is ensured as the respondents directly send the filled up questionnaire to the researcher.

Limitations of Questionnaire Method

1. It is always found that the response rate in questionnaire is very poor compared to using schedules.
2. Bias of the respondent cannot be determined easily.
3. Only if the respondents are educated, questionnaire could be used for collecting information
4. Follow up of non - response or unfilled questionnaire only adds to the cost and time.
5. There is no scope for giving any clarification to the respondents in case of need.
6. Accuracy of response cannot be ensured.
7. A lot of care is required to design and structure a questionnaire. Hence, unless a scholar is good at drafting a questionnaire, this method cannot be adopted for data collection.

Differences between Questionnaire and Schedule

BASIS OF DIFFERENCE	QUESTIONNAIRE	SCHEDULE
Usage	Respondent himself records the answers obtained	Researcher or his trained representative records the answers
Cost	Relatively cheaper as it is sent by mail to the targeted respondents	Relatively costly as it is recorded with personal presence
Coverage	Wide coverage possible, as it can be sent to any place by post	Very expansive for wide coverage
Degree of Response	Degree of response is low	Degree of response is high
Quality of Response	Not good, as the respondent answers the questions the way it is understood	Skill of researcher persuades the respondent to respond effectively
Identification of Respondent	It is not known clearly who answered the questionnaire and this in turn might affect accuracy of information obtained	More accurate information can be sought

Prerequisites of Questionnaire

1. The scope of the survey should be clearly determined in advance. This will help to determine the segment of informants or respondents from whom the information should be collected
2. The purpose of the survey must be established and stated in the questionnaire
3. The researcher should prepare the draft questionnaire and revise it until it is fit for the purpose of enquiry.
4. Follow up arrangement after dispatching the questionnaire should be planned in advance.
5. Maintenance of list of respondents, to whom the questionnaire is sent, is necessary.
6. Whether to use structured or unstructured questionnaire is an important decision to be taken.
7. Time frame for different activities connected with the questionnaire like, preparation of questionnaire, its finalisation, printing, collecting list of respondents and their addresses, despatch and follow up, etc.

Constructing Questionnaire

1. Questionnaire should be printed / cyclostyled / xeroxed
2. The first part of the questionnaire should specify the object or purpose for which the information is required.
3. An assurance to the respondent that the information furnished would be kept confidentially must be given.
4. Some introduction about the person / organisation who is collecting the information is to be given. Sometimes, this is given as a letter enclosed with the questionnaire.
5. Questions should be constructed using simple language and technical jargons, terms, concepts should all be avoided.
6. Questions should never be lengthy. It is wiser not to combine a number of questions in one question itself.
7. Each question should be specific and clear.
8. Personal questions on wealth, habits, etc., could be avoided.
9. Questions should be given in a sequence. [For example, Question relating to marital status should precede the question about the number of children]
10. Questions should not require any referencing before replying.
11. Questions should not force the respondent to recall from his memory anything to answer.

12. Questions needing computation / calculation / consultation should be avoided.
13. Questions on sentiments / belief / faith should be avoided.
14. Repetition of question should be eliminated.
15. In case of any instruction is required for filling up the questionnaire, such instructions should be given separately specifying the question number and the related instructions. For example, the Application form for University Examination, is given with instructions sheet regarding fees to be paid, mode of payment, subject code, etc.
16. Sufficient space should be given for answering questions
17. The questionnaire should be made attractive, as otherwise, the respondents may not be motivated to answer.
18. If any diagramme or map is used, then it should be printed clearly.
19. Questions, which, cross check the response, could be built in to the questionnaire.
20. Instructions regarding how to return the filled up questionnaire must be given. It is desirable that a self-addressed sufficiently stamped envelope is sent along with the questionnaire to enable the respondents to send the filled up questionnaire.

Steps involved in preparing questionnaire

1. Define the Purpose and Objectives

- Clearly identify what information you need to collect and why. This involves reviewing your research objectives or project goals to ensure each question serves a specific purpose

2. Identify the Target Audience

- Determine who will be answering the questionnaire. Understanding the demographics, language proficiency, and characteristics of your respondents helps tailor the questions appropriately.

3. Choose the Method of Distribution

- Decide how you will reach your respondents (e.g., online, face-to-face, telephone, mail), as this can influence question format and length.

4. Draft a Comprehensive List of Questions

- List all potential questions that could help achieve your objectives. At this stage, focus on content rather than wording or order.

5. Refine Question Wording and Format

- Develop clear, concise, and unbiased questions. Choose between open-ended and closed-ended formats as appropriate. Ensure questions are easy to understand and avoid ambiguity, leading, or sensitive questions unless necessary.

6. Organize Questions in a Logical Sequence

- Arrange questions to create a natural flow, typically starting with simple or general questions and progressing to more complex or sensitive ones. This helps keep respondents engaged and improves data quality.

7. Decide on Response Formats

- For each question, determine the type of response required (e.g., multiple choice, Likert scale, short answer) and ensure response options are exhaustive and mutually exclusive where applicable.

8. Check Questionnaire Length and Layout

- Ensure the questionnaire is not too long, as this can reduce response rates. Format the questionnaire for clarity, with clear instructions and an appealing, easy-to-follow layout.

9. Pretest (Pilot) the Questionnaire

- Test the questionnaire on a small sample from your target audience to identify confusing questions, layout issues, or other problems. Use feedback to make necessary revisions.

10. Finalize and Distribute

- Make final adjustments based on pilot feedback. Ensure the questionnaire is ready for distribution and then administer it to your target population.

Interview Techniques

Discussion Points

1. Meaning& role,
2. Objectives ,
3. Types,
4. The process,
5. Advantages and Disadvantages

Meaning of Interview

One of the very old methods of compiling data [facts] is the interview method. Interview method involves direct or indirect meeting of the respondent by the researcher. The researcher determines the questions to be raised at the time of interview and then elicit the response for them. The reply given is either written down in a note - book or recorded in audio or video cassette. This method has to be necessarily adopted whenever details regarding any confidential matter are to be collected or when authentic information about any thing is to be obtained or the research requires data collection directly from the respondents or there is no other way of collecting data.

OBJECTIVES OF INTERVIEW

- 1. Create relation between interviewer and interviewee**
- 2. Gain inner feelings of the respondent**
- 3. Collect information about an existing social problem**
- 4. Create a source of knowledge**
- 5. Provide rich hypothesis**
- 6. Observe the situation quickly**
- 7. Collect additional information**
- 8. Reduce the distance between human beings**
- 9. Draw quantitative facts**
- 10. Gain new knowledge about an area**



Types of Interviews

1. **Direct interviews** : The **direct interview method** is a real-time, face-to-face (or synchronous virtual) conversation where an interviewer asks questions directly to a subject or candidate. It is widely used in **qualitative research, journalism, and job recruitment** to gather first-hand insights, attitudes, and behavior

a. **Structured Interview**: It uses a fixed set of standardized questions asked in the exact same order for every candidate.

b. **Unstructured Interview**: Unstructured interviews are informal and exploratory. There are no set questions, and the direction of the interview is driven by the candidate's previous answers and the natural flow of the conversation

2. **Indirect interviews**: The indirect interview method (often called **indirect oral investigation**) is a technique where an investigator collects information by questioning third parties rather than the primary subject. It is primarily used when the main subject is unreachable, hesitant, or likely to provide biased or inaccurate information

Merits of Interview method

1. Interview enables personal contact with which factual information could be obtained.
2. The interviewer can guide the respondents to understand the questions and suitably rephrase or modify the questions.
3. The non-response in this category of data collection is low.
4. Spontaneous reply of the respondents can be obtained which would truly reflect their opinion.
5. Interviewer has the flexibility to handle the interview to collect relevant information.
6. The interviewer can combine observation with interview so that a lot of additional information about the environment, nature, behaviour and attitude of the respondent could also be noted.
7. The interviewer can be persuasive to obtain the response of the respondent while this may not be possible in the case of questionnaire method of data collection.
8. The interviewer can adopt himself better by using the language with which he can reach the respondents better and the educational level of the respondent.
9. It is also possible to obtain personal information which will not be revealed through questionnaire.

Demerits of Interview Method

1. This method presupposes the existence of rapport between the respondent and the interviewer. In case there is no rapport, the interview cannot be held at all.
2. It is a costly method of data collection, as the interviewers should be trained, oriented and supported.
3. The time taken for covering a large sample of respondents is more than what it is under questionnaire method.
4. It is possible for a respondent to conceal his real opinion or views and so genuine data may not be available.
5. The interviewer may also be biased and thereby his bias may influence the interview outcome.
6. Interview method requires an elaborate organisational arrangement.
7. Sometimes this method cannot be followed as it would require fixing up appointment, specifying the questions that are to be answered, duration of interview, the person interviewing, the organizational background, the purpose of interview, etc., In spite of providing all this information, it may not be possible to get an appointment to conduct the interview.

Observation

Presentation by,
Prof. Basavaraj S.K

Discussion points

- Meaning & characteristics,
- Types,
- Advantages & disadvantages

Meaning

Observation may be defined as, "sensible application of sense organs in understanding less explained or unexplained phenomena." Using the sense organs, one could see and understand things. One could listen to what is being explained to understand anything. One may also understand things using the power to smell or feeling of touch. It is also possible by interacting with people to understand anything. Whenever a researcher is unable to compile information through any other method, then he has to effectively apply his sense organs to observe and explain. So it may be said that observation involves recording of information applying visual understanding backed by alert sense organs.

Types of observations

1. Structured observation : When observation takes place strictly in accordance with a plan or a design prepared in advance, it is called structured observation.
2. Unstructured observation : In this type of observation, there is no advance designing of what, how, when , who, etc., of observation. The observer is given the freedom to decide on the spot, to observe everything that is relevant.
3. Participant observation : In this method, the observer is very much present in the midst of what is observed. For example, suppose a researcher is studying the life style of a hill tribe, then he might understand the life style of the tribe better only when he stays with them.
4. Non-participant observation : This is a method in which the observer remain detached from whatever is happening around and does not involve himself in any activities taking place. He is present only to observe and not to take part in the activities.

5. Controlled observation : This is a slightly modified version of structured observation. In this case, the observer performs his work in an environment or situation, which is very much planned or designed or set. For example, sometimes to test the effectiveness and alertness of airport security system, mock event [like hijacking or fire accident] is carried out.
6. Uncontrolled observation : This method is a modified version of unstructured observation. No attempt is made to intervene the naturalness of the environment or situation. The observer is at freedom to observe whatever is taking place around him in the natural set up. For example, an animal lover may like to watch the conduct and behavior of the animals in their natural habitat.

Merits of observation method of data collection(Advantages)

1. If observation is done correctly, the scope for bias is very much minimized.
2. Through observation, the current scenario in which anything is happening is noticed and explained. There is no interpretation of how things would have happened in the past or will happen in future, etc.
3. As there is no need to get any reply or details from the respondents, observation does not require any cooperation of the respondents.
4. This is a fairly reliable method, provided the observer is well experienced, trained and sincere.
5. Whenever respondents are illiterate or incapable of answering any question, [due to language barrier or cultural background, etc], observation is the only method of data collection available.

Limitations (Disadvantages)

1. This is a relatively costly method of data collection, because the observer has to be present on the spot to observe whatever is happening. Further training an observer is the additional cost to be incurred.
2. The scope for the bias of the observer interfering in what is observed and understood is high.
3. It could be noticed that what is observed may bring out only part of the facts, while data collected through questionnaire or interview ensure better coverage.
4. Sometimes, it may not be possible to observe what is targeted. This may be due to the difficulty in reaching such people or spot.
5. There is a lot of scope for the observer to get distracted or influenced by unexpected factors which would affect the accuracy of information collected.
6. A very important limitation is that only the observer can effectively establish link among whatever he has observed to give a meaningful interpretation. Based on his report nobody else can give satisfactory interpretation. Therefore, the ability to relate things and interpret them intelligently is a quality expected of an observer.

Secondary Data

Secondary data refers to the information or facts already collected. Such data are collected with the objective of understanding the past status of any variable. Or the data collected and reported by some source is accessed and used for the objective of a study. Normally in research, the scholars collect published data, analyse them, to explain the relationship between variables, which may not have been studied or the relationship might be explained in a new way. For example, data relating to production and productivity of rice in Tamil Nadu may be analysed by different scholars for different purposes.

Advantages

1. Research is more accurate.
2. Past data is having record.
3. Third party can test the accuracy of research.
4. Evidences prove the research hypothesis.

Disadvantages

1. Not suitable for all types of researches.
2. Requires authentic sources.
3. No personal inquiry is possible.

Sources of Secondary Data

1. Books.
2. Periodicals.
3. Research thesis and dissertation.
4. Bibliographies.
5. Encyclopedia.
6. Statistical data source.
7. Directories and year book.

How to give citations

1. Referencing a book

- **Basic format to reference a book**

- Author or authors. The surname is followed by first initials.
- Year of publication of the book (in round brackets).
- Book title (in italics).
- Edition (in round brackets), if other than first.
- Publisher.

Author



Year



Title in *italics*



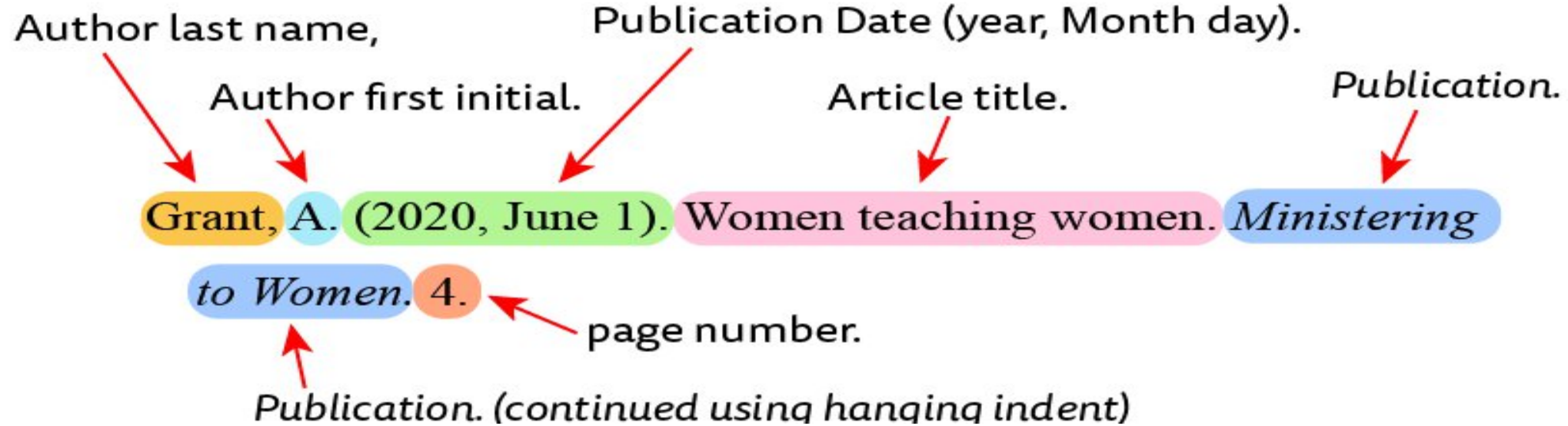
Publisher



Hasler, E. (2018). *The built environment*. Liverpool University Press.

2. Periodical referencing

APA Print Magazine Citation



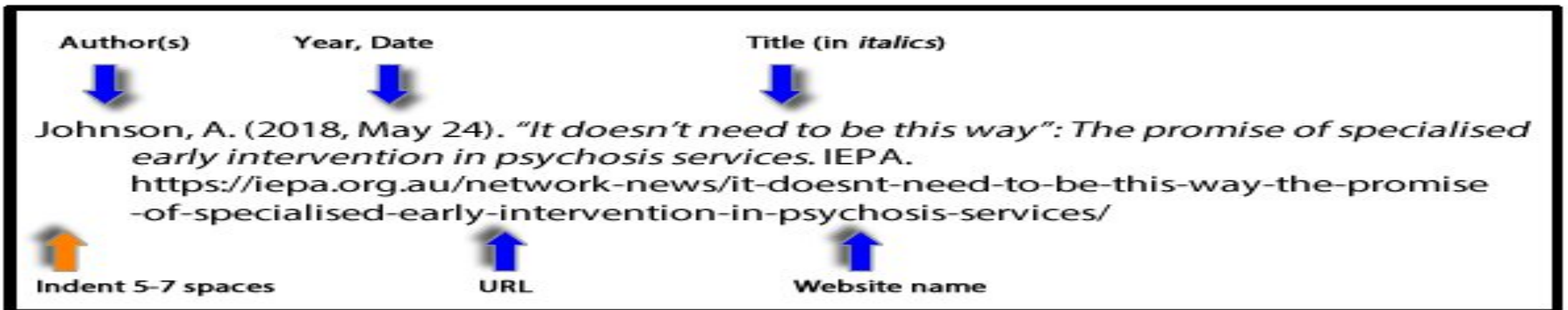
Referencing Research thesis / dissertation



Referencing Encyclopedia.



Referencing websites



Selection of Appropriate Method of Data Collection

1. Nature and scope and objective of the research.
2. Budget Constrain.
3. Availability of Time.
4. Degree of Accuracy required.
5. Period of study.

- **Measurement and Scaling Techniques:**
Discussion Points.

- A. Basic measurement scales-

1. Nominal scale,
2. Ordinal scale,
3. Interval scale,
4. Ratio scale.

- B. Attitude measurement scale –

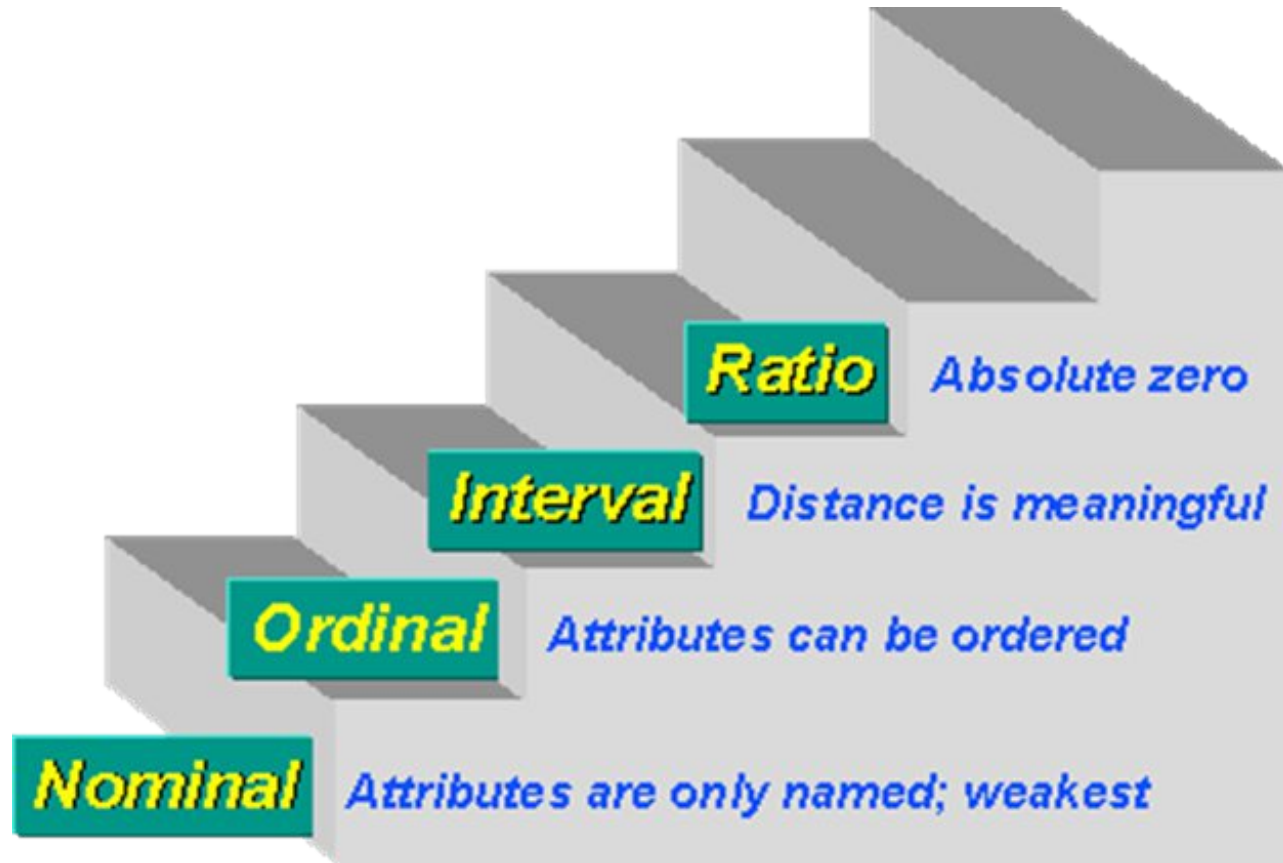
1. Likert Scale,
2. Semantic Differential Scale,
3. Thurston scale,

Scales of Measurement

The process of assigning numbers to objects in such a way that specific properties of the objects are faithfully represented by specific properties of the numbers. Such ways of assigning numbers do not attempt to measure the total phenomenon, but only a specific set of attributes.

A. Basic measurement scales:

1. Nominal scales
2. Ordinal scales
3. Interval scales
4. Ratio scales



1. Nominal Scale:

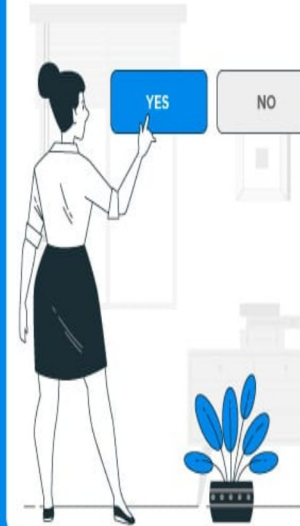
A nominal scale of measurement deals with variables that are non-numeric or where the numbers have no value. The lowest measurement level you can use, from a statistical point of view, is a nominal scale. A nominal scale, as the name implies, is simply some placing of data into categories, without any order or structure.

A physical example of a nominal scale is the terms we use for colors. The underlying spectrum is ordered but the names are nominal. In research activities a YES/NO scale is nominal. It has no order and there is no distance between YES and NO.

Characteristics of Nominal Scale

1

Divides a variable in 2 or more categories. Examples are agree/disagree, yes/no, etc.



2

Is qualitative in nature. Numbers are used only to categorize or identify objects.



3

Numbers don't define object-related characteristics. Each number is assigned to a random object.

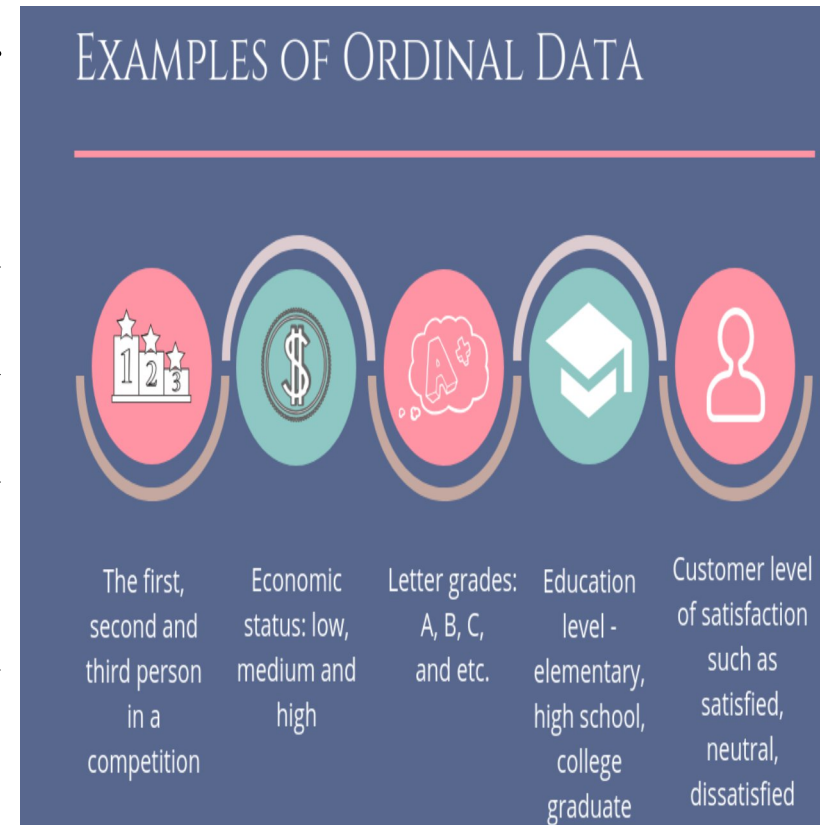


2. Ordinal Scale:

The ordinal type allows for rank order (1st, 2nd, 3rd, etc.) by which data can be sorted, but still does not allow for relative degree of difference between them.

An ordinal scale of measurement looks at variables where the order matters but the differences do not matter. When you think of 'ordinal,' think of the word 'order.' In the case of letter grades, we don't really know how much better an A is than a D. We know that A is better than B, which is better than C, and so on. But is A four times better than D? Is it two times better?

In this case, the order is important but not the differences. Examples include, on one hand, dichotomous data with dichotomous (or dichotomized) values such as 'sick' vs. 'healthy' when measuring health, 'guilty' vs. 'innocent' when making judgments in courts, 'wrong/false' vs. 'right/true' when measuring truth value, and, on the other hand, non-dichotomous data consisting of a spectrum of values, such as 'completely agree', 'mostly agree', 'mostly disagree', 'completely disagree' when measuring opinion.



3. Interval Scale:

Interval scales are numeric scales in which we know not only the order, but also the exact differences between the values.

The classic example of an interval scale is Celsius temperature because the difference between each value is the same. For example, the difference between 60 and 50 degrees is a measurable 10 degrees, as is the difference between 80 and 70 degrees. Time is another good example of an interval scale in which the increments are known, consistent, and measurable



4. The ratio scale

The ratio scale of measurement is the most informative scale. It is an interval scale with the additional property that its zero position indicates the absence of the quantity being measured. You can think of a ratio scale as the three earlier scales rolled up in one. Like a nominal scale, it provides a name or category for each object (the numbers serve as labels). Like an ordinal scale, the objects are ordered (in terms of the ordering of the numbers). Like an interval scale, the same difference at two places on the scale has the same meaning. And in addition, the same ratio at two places on the scale also carries the same meaning.

Ratio scales are the ultimate nirvana when it comes to measurement scales because they tell us about the order, they tell us the exact value between units and they also have an absolute zero—which allows for a wide range of both descriptive and inferential statistics to be applied. Everything above about interval data applies to ratio scales i.e. ratio scales have a clear definition of zero.

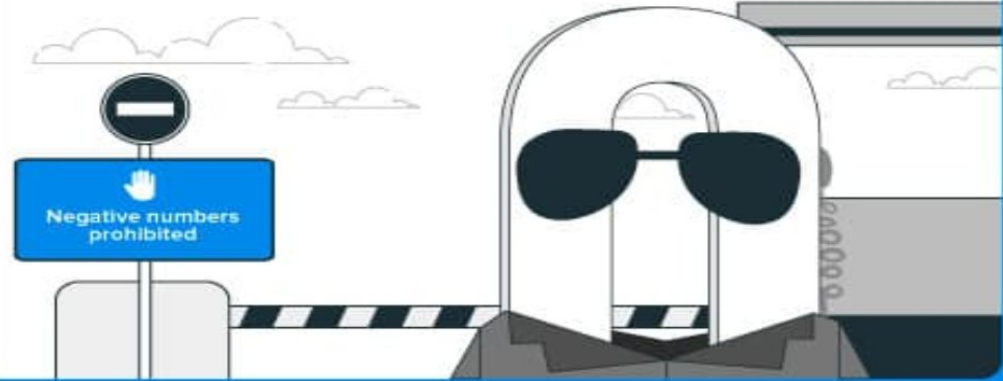
Good examples of ratio variables include height and weight. Ratio scales provide a wealth of possibilities when it comes to statistical analysis. These variables can be meaningfully added, subtracted, multiplied, divided (ratios). Central tendency can be measured by mode, median, or mean; measures of dispersion, such as standard deviation and coefficient of variation can also be calculated from ratio scales.

Characteristics of Ratio Scale

Has an absolute zero characteristic. It has orders and equally distanced value between units.



Ratio scale data does not have any negative numerical value. For example, weight cannot be negative, -20 Kgs does not exist.



The data values of a ratio scale can be added, subtracted, multiplied and divided. All statistical analysis can be calculated using ratio scale.



It has ratio scale units which have several unique and useful properties. One of them is they allow unit conversion.



Scale	Meaning	Characteristics	Management Examples
Nominal Scale	Used for naming or categorizing	No order, no ranking	Gender (Male/Female), Department (HR, Finance, Marketing), Marital Status, Type of Bank Account, Brand Preference (Samsung, Apple, Vivo)
Ordinal Scale	Used for ranking or ordering	Order exists, but differences are not equal	Customer Satisfaction (Very Dissatisfied to Very Satisfied), Employee Performance Rating (Poor, Average, Good, Excellent), Class Rank, Socio-economic Status (Low, Middle, High)
Interval Scale	Ordered with equal intervals, but no true zero	Addition and subtraction possible	Temperature (°C, °F), IQ Score, Calendar Years (2023, 2024, 2025), Employee Attitude Scores from standardized scales
Ratio Scale	Ordered with equal intervals and a true zero	All mathematical operations possible	Salary, Sales Revenue, Profit, Number of Employees, Age, Working Hours, Production Units, Market Share (%)

Questionnaire based on scales

1. **Nominal Scale Question:**

Which specialization have you chosen?

- Finance
- Marketing
- HR
- Operations

☞ Only classification; no ranking.

3. **Interval Scale Question:**

Office temperature today:

- 20°C
- 25°C
- 30°C

☞ Difference between 20°C and 25°C is same as between 25°C and 30°C, but 0°C does not mean "no temperature."

2. **Ordinal Scale Question:**

Rate the quality of college facilities.

- Poor
- Fair
- Good
- Very Good
- Excellent

☞ Rankings exist, but the gap between Good and Very Good is not exactly measurable.

4. **Ratio Scale Question:**

Monthly salary of employees.

- ₹20,000
- ₹40,000
- ₹60,000

☞ ₹40,000 is exactly twice ₹20,000, and ₹0 means no salary.

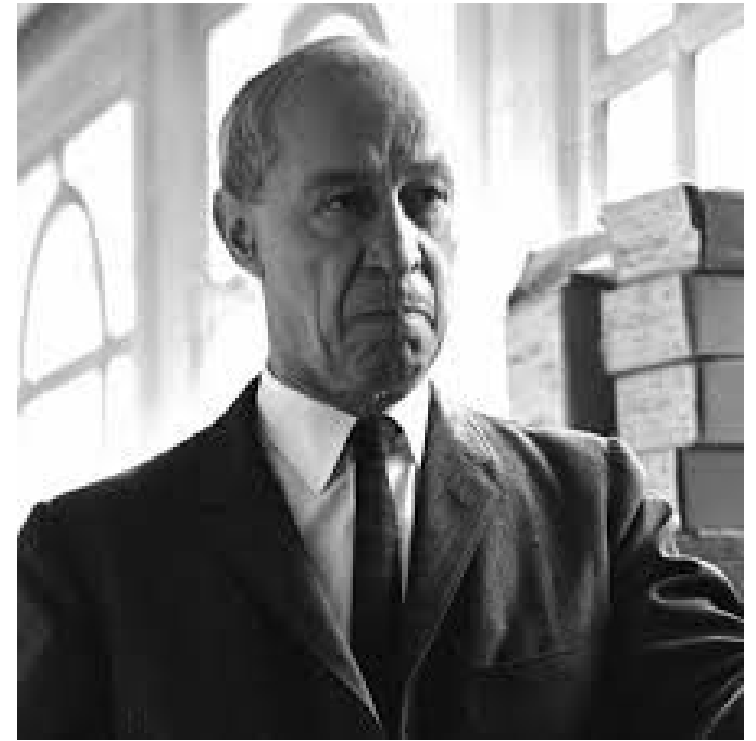
B. Attitude Measurement scales.

Attitude measurement scale –

1. Likert's Scale
2. Semantic Differential Scale
3. Thurston scale,

What is Likert scale?

- The Likert scale was invented by American social psychologist **Rensis Likert**. He developed the psychometric scale in 1932 as part of his doctoral dissertation at Columbia University.
- A **Likert scale** is a psychometric rating scale commonly used in surveys to measure people's attitudes, opinions, or behaviors. Instead of a simple "yes" or "no," respondents choose from a graduated range of options (e.g., from "strongly disagree" to "strongly agree") to indicate the intensity of their feelings.



Rensis Likert 1903-1981

1. Likert's scale

A Likert scale is a unidimensional scale that researchers use to collect respondents' attitudes and opinions. Researchers often use this psychometric scale to understand the views and perspectives towards a brand, product, or target market.

- **Types of Likert Scales**

- The **Likert Scale** has become a favorite among researchers to collect opinions about customer satisfaction or employee experience. You can divide this scale primarily into two major types:
- Even Likert Scale.
- Odd Likert Scale.

a. Even Likert Scale

Researchers use even Likert scales to collect extreme feedback without providing a neutral option.

4-Point Likert Scale for importance: This type of Likert scale allows researchers to include four extreme options without a neutral choice. Here the various degrees of importance are represented in a 4-Point Likert Scale.

8-Point Likelihood of recommendation: This is a variation of the previously explained 4-point Likert scale, the only difference being, this scale has eight options to collect feedback about the likelihood of a recommendation.

4 Point Scale

Four-point Likert scale example for agreement

The ambiance of the hotel is most attractive.

- Strongly agree
- Agree
- Disagree
- Strongly disagree

For Satisfaction

How are satisfied with the hotel food quality?

- Very satisfied
- Satisfied
- Dissatisfied
- Very dissatisfied

For Rating

How would you rate the staff friendliness?

- Excellent
- Good
- Better
- Poor

For Likelihood

How likely are you to recommend this hotel?

- Very unlikely
- Unlikely
- Likely
- Very likely

For Frequency

How frequently do you visit our hotel?

- Never
- Rarely
- Often
- Every Time

b. Odd Likert Scales

Odd Likert Scale

Researchers use the odd Likert scale to give the respondents the choice of responding neutrally.

5-point Likert scale: With five answer options, researchers use this odd Likert scale question to gather information about a topic by including a neutral answer option for respondents to select if they don't wish to answer from the extreme choices in their research design.

7-point Likert scale: The 7-point Likert scale adds two more answer options at extreme ends of a 5-point Likert scale question.

9-point Likert scale: A 9-point Likert scale is quite uncommon, but you can use it by adding two more answer options to the 7-point Likert scale question.

5 Point Scales

Satisfaction	Likelihood	Level of concern
1. Very dissatisfied 2. Dissatisfied 3. Neither dissatisfied or satisfied 4. Satisfied 5. Very satisfied	1. Very unlikely 2. Unlikely 3. Neutral 4. Likely 5. Very likely	1. Very unconcerned 2. Unconcerned 3. Neutral 4. Concerned 5. Very concerned

Agreement	Frequency	Awareness
1. Strongly disagree 2. Disagree 3. Neither agree or disagree 4. Agree 5. Strongly agree	1. Never 2. Rarely 3. Sometimes 4. Often 5. Always	1. Very unaware 2. Unaware 3. Neither aware or unaware 4. ware 5. Very aware

Familiarity	Quality	Importance
1. Very unfamiliar 2. Unfamiliar 3. Somewhat familiar 4. Familiar 5. Very familiar	1. Very poor 2. Poor 3. Acceptable 4. Good 5. Very good	1. Very unimportant 2. Unimportant 3. Neutral 4. Important 5. Very important

7 Points Likerts Scales

7-point scales

Agreement	Frequency	Appropriateness
1. Strongly disagree 2. Disagree 3. Somewhat disagree 4. Neither agree or disagree 5. Somewhat agree 6. Agree 7. Strongly agree	1. Never 2. Rarely (less than 10% of the time) 3. Occasionally (about 30% of the time) 4. Sometimes (about 50% of the time) 5. Frequently (about 70% of time) 6. Usually (about 90% of the time) 7. Every time	1. Absolutely inappropriate 2. Inappropriate 3. Slightly inappropriate 4. Neutral 5. Slightly appropriate 6. Appropriate 7. Absolutely appropriate
Satisfaction	Reflective of me	Level of difficulty
1. Very dissatisfied 2. Dissatisfied 3. Slightly dissatisfied 4. Neutral 5. Slightly satisfied 6. Satisfied 7. Very satisfied	1. Very untrue of me 2. Untrue of me 3. Somewhat untrue of me 4. Neutral 5. Somewhat true of me 6. True of me 7. Very true of me	1. Very easy 2. Easy 3. Somewhat easy 4. Neutral 5. Somewhat hard 6. Hard 7. Very hard
Priority	Quality	Importance
1. Not a priority 2. Low priority 3. Somewhat a priority 4. Neutral 5. Moderate priority 6. High priority 7. Essential priority	1. Very poor 2. Poor 3. Below average 4. Average 5. Above Average 6. Good 7. Excellent	1. Very unimportant 2. Unimportant 3. Slightly unimportant 4. Neutral 5. Slightly important 6. Important 7. Very important

2. The semantic differential scale

The semantic differential scale is used for measuring the meaning of things and concepts (Snider & Osgood, 1969). Participants indicate where they feel a product or experience lies on a 7-point scale between polar adjectives (e.g., strong-weak, good-bad).

Unpleasant	1	2	3	4	5	6	7	Pleasant
Rude	1	2	3	4	5	6	7	Polite
Cold	1	2	3	4	5	6	7	Warm
Callous	1	2	3	4	5	6	7	Considerate
Inhuman	1	2	3	4	5	6	7	Human
Noncommunicative	1	2	3	4	5	6	7	Communicative
Disrespecting	1	2	3	4	5	6	7	Respectful
Insincere	1	2	3	4	5	6	7	Friendly
Unconcern	1	2	3	4	5	6	7	Empathy

Example of Semantic Differential Scale

Unpleasant	1	2	3	4	5	6	7	Pleasant
Rude	1	2	3	4	5	6	7	Polite
Cold	1	2	3	4	5	6	7	Warm
Callous	1	2	3	4	5	6	7	Considerate
Inhuman	1	2	3	4	5	6	7	Human
Noncommunicative	1	2	3	4	5	6	7	Communicative
Disrespecting	1	2	3	4	5	6	7	Respectful
Insincere	1	2	3	4	5	6	7	Friendly
Unconcern	1	2	3	4	5	6	7	Empathy

Thurstone Scale

- The Thurstone scale is a psychometric measurement tool developed by Louis Leon Thurstone in 1928. It's sometimes seen as a forerunner to the Likert scales as both of them are made of statements about a topic.
- But in the Thurstone scale, respondents reply to a list of related agree-disagree-neutral statements of varying weights. The respondent's attitude is calculated as the sum total of the statements they agree with.
- Unlike Likert scales which only allow respondents to provide one answer per question, Thurston scales allow them to give multiple answers by assigning each item on the list a score (usually 1-11). This allows researchers to get more detailed results from their surveys.

Steps involved in developing the Thurstone Scale.

- **Step 1:** Pin down your research question. It should be related to attitude, and should focus on a single topic, like “end-of-life care” or “performance related pay”.
- **Step 2:** Create a list of statements that agree, disagree with or are neutral about the topic to varying degrees. The more statements you come up with, the more useful your data will be.
- **Step 3:** Assign a score to each statement from 1-11 based on how supportive you believe the statement is. A higher score equals a more supportive stance while a lower score equals a more unfavorable stance. The scores are usually assigned in consensus with a panel of experts to reduce bias.
- **Step 4:** Each member of the panel should assign an individual score to each statement. Based on the mean of these scores, each statement is assigned an overall score and sorted in ascending order.
- **Step 5:** Once all of your questions have been created, you will then need to conduct pilot interviews with potential respondents. This helps determine whether or not your questions are clear and easy enough for them to understand.

Example of developing Thurstone scale

- For example, you are creating a Thurston questionnaire on employee attitudes toward performance-related pay (PRP). Your statements may include:
 - 1.PRP gives an incentive to go above and beyond job expectations.
 - 2.Good performers are rewarded fast due to PRP.
 - 3.PRP makes team members less willing to help out.
 - 4.PRP increases the quality of the work.
 - 5.Management uses PRP to unjustly reward favorites.

Suppose there are 10 judges giving their opinions

Questions/ Options	1	2	3	4	5
1	1	1	1	1	1
2	1	1	5	1	2
3 1st Quortile	6 (Mean= 2)	5 (Mean=2)	1 (Mean =2)	0 (Mean= 0.4)	2 (Mean=1.4)
4	0	1	2	0	1
5	2	2	1	0	1
6 (Median	0	0	0	0	0
7	0	0	0	0	1
8	0	0	0	6	1
9 3 rd quortile	0 (Mean=0)	0 (Mean=0)	0 (Mean=0)	2 (Mean=2)	1 (Mean= 0.6)
10	0	0	0	2	0
11	0	0	0	0	0
Total	10	10	10	10	10

Advantages

- You can write as many statements as you want in the Thurston scale questionnaire. In fact, the more statements you have, the richer your data will be.
- The Thurston scale is more reliable because it's designed by experts.
- The Thurston scale lets you assign an average score for each respondent. This lets you compare the results of different respondents and make predictions.

Disadvantages

- The Thurston scale can be time consuming, in terms of both designing the surveys and analyzing the results.
- Thurstone scales are complex to execute and will require you to do some mathematical calculations. If you just want to quickly create some questions and share the survey, this might not be the right tool for you.

Multidimensional Scaling Techniques

Multidimensional Scaling (MDS) is a class of procedures for representing perception and preferences of respondents spatially by means of a visual display. Perceived or psychological relationships among points in a multidimensional space. These geometric representations are often called spatial maps. The axes of the spatial map are assumed to denote the psychological bases or underlying dimensions respondents use to form perceptions and preferences for stimuli. MDS has been used in marketing to identify:

1. The number and nature of dimensions consumers use to perceive different brands in the marketplace
2. The positioning of current brands on these dimensions
3. The positioning of consumers ideal brand on these dimensions

Application of Multi Dimensional scaling

- Image measurement. Compare the customers' and noncustomers' perceptions of the firm with the firm's perceptions of itself and thus identify perceptual gaps.
- Market segmentation. Position brands and consumers in the same space and thus identify groups of consumers with relatively homogeneous perceptions.
- New product development. To look for gaps in the spatial map, which indicate potential opportunities for positioning new products. Also, to evaluate new product concepts and existing brands on attest basis to determine how consumers perceive the new concepts. The proportion of preferences for each new product is one indicator of its success.
- Assessing advertising effectiveness. Spatial maps can be used to determine whether advertising has been successful in achieving the desired brand positioning.
- Pricing analysis. Spatial maps developed with and without pricing information can be compared to determine the impact of pricing.  Channel decisions. Judgments on compatibility of brands with different retail outlets could lead to spatial maps useful for making channel decisions.
- Attitude scale construction. MDS techniques can be used to develop the appropriate dimensionality and configuration of the attitude space.

Example of multi dimensional Scaling

		LIKELIHOOD			
		10	20	30	40
		Low	Medium	High	Very High
IMPACT	10 Low	4	1	1	3
	20 Medium	4	5	7	7
	30 High	7	7	10	6
	40 Very High	7	12	14	9



Practical example

- **Research Problem**
- A smartphone company wants to know how customers perceive different smartphone brands.
- Suppose five brands are selected:
 - Samsung
 - Apple
 - OnePlus
 - Vivo
 - Redmi

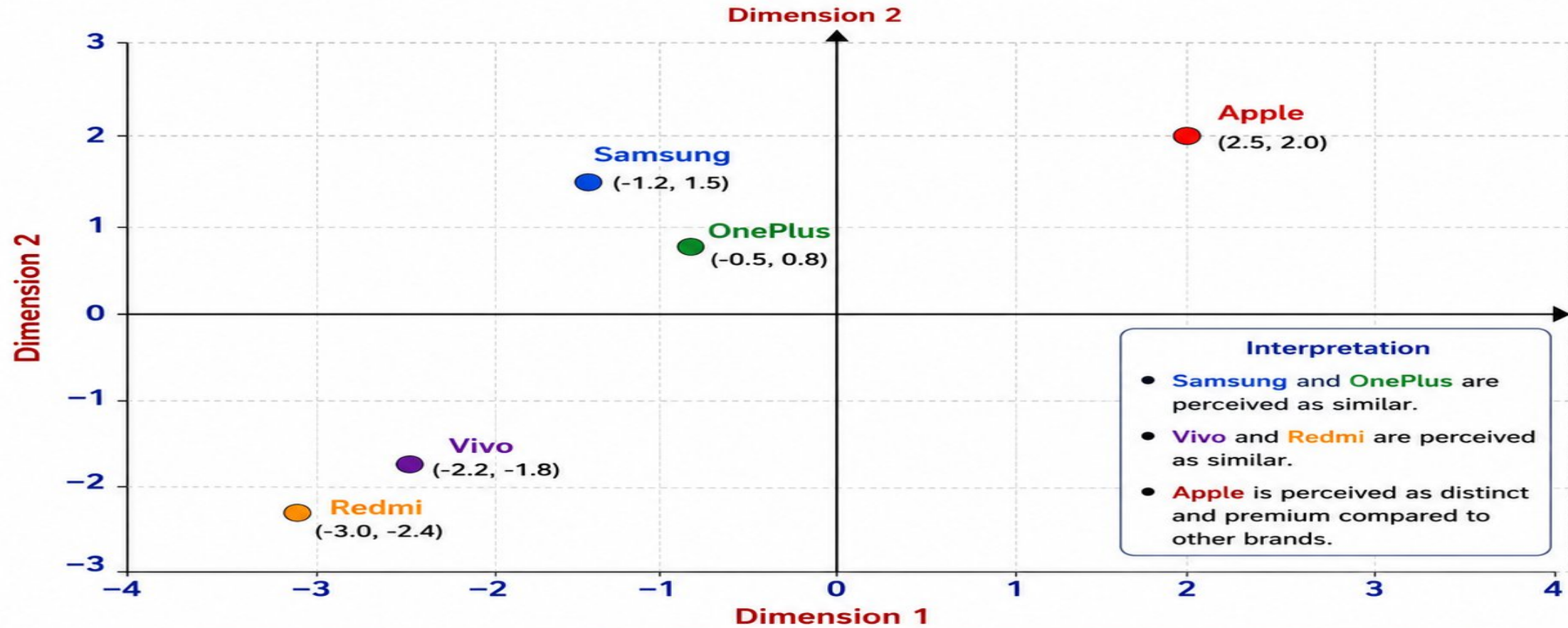
- Assume 50 respondents rated the similarity between every pair of brands on a scale of
- **1 = Very Similar**
- **10 = Very Different**
- The average ratings are shown below.

Survey result

Brand	Samsung	Apple	OnePlus	Vivo	Redmi
Samsung	0	6	3	5	4
Apple	6	0	5	8	9
OnePlus	3	5	0	4	5
Vivo	5	8	4	0	2
Redmi	4	9	5	2	0

Multidimensional Scaling (MDS) – Perceptual Map of Smartphone Brands

Based on Customer Perceptions of Similarity/Dissimilarity



Coordinates from MDS Solution		
Brand	Dimension 1	Dimension 2
● Samsung	-1.2	1.5
● Apple	2.5	2.0
● OnePlus	-0.5	0.8
● Vivo	-2.2	-1.8
● Redmi	-3.0	-2.4

Model Summary (Goodness of Fit)	
Statistic	Value
Stress Value	0.08
RSQ (Squared Correlation)	0.95
Interpretation	Good Fit

Stress Value Interpretation

< 0.05	Excellent
0.05 – 0.10	Good
0.10 – 0.20	Acceptable
> 0.20	Poor

← Our Stress = 0.08
(Good Fit)

Advantages of Multidimensional Scaling (MDS)

- Hints at possible solutions rather than prescribes. Tells you the possibilities, leaves specifics of solution to you.
- Dimensions (axis) can suggest facets.
- Similarity maps are easy to understand
- Helps identify what dimensions are important
 - cluster analysis can be done on results of MDS. This gives cleaner results.
- Cluster Analysis and MDS are complementary



Limitations of MDS

- Perceptual mapping has not been shown to be reliable across different methods
- The effect of market events on perceptual maps cannot be ascertained
- The interpretation of dimensions is difficult
- When more than two or three dimensions are needed, usefulness is reduced

Unit-5

Data Analysis and Report Writing

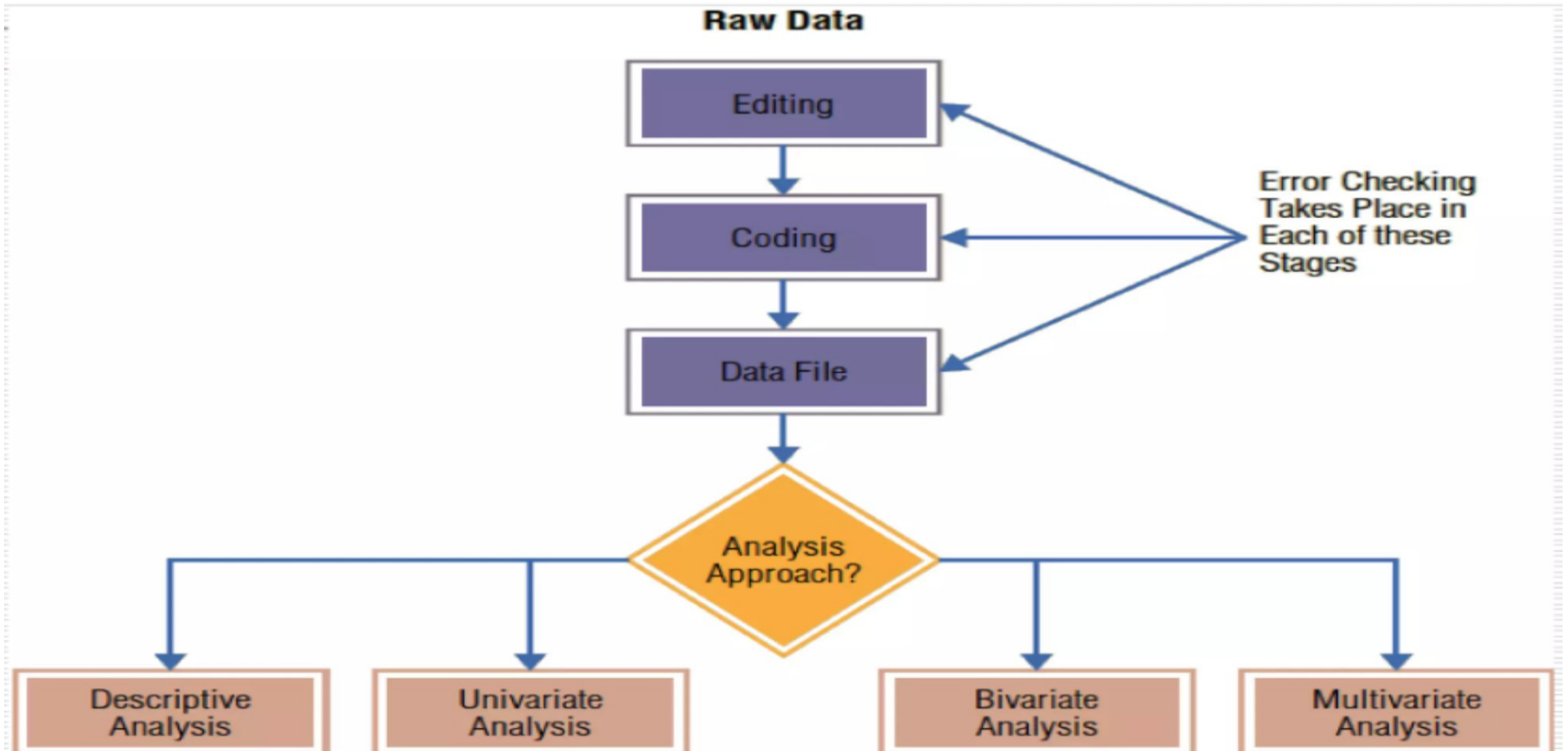
Presentation by
Dr. Basavaraj S.Kudachimath
VTU Belagavi

Data Analysis and Report Writing

Discussion Points:

- Editing, Coding, Classification, Tabulation, Validation.
- Analysis and Interpretation, Report writing and presentation of results, Importance of report writing,
- types of research reports, Report structure, Guidelines for effective documentation.

Overview of Stages in Data Analysis



What is Editing?

- The process of checking and adjusting responses in the completed questionnaires for omissions, legibility, and consistency and readying them for coding and storage.
- Detects errors and omissions, corrects them when possible, and certifies that minimum data quality standards are achieved

Editing of data ensures following

1. Complete the data collection
2. Consistency in collecting data
3. Accurate data
4. Adequate data
5. Homogeneous data

Types of Editing

1. Field Editing

Preliminary editing by a field supervisor on the same day as the interview to catch technical omissions, check legibility of handwriting, and clarify responses that are logically or conceptually inconsistent.

2. In-house or Central Editing

Editing performed by a central office staff; often done more rigorously than field editing

Problems/ Pitfalls in Editing

- Allowing subjectivity to enter into the editing process.
- Data editors should be intelligent, experienced, and objective.
- Failing to have a systematic procedure for assessing the questionnaires developed by the research analyst
- An editor should have clearly defined decision rules to follow.

Coding

- Code is a numerical score or symbol.
- Coding is the process of identifying and classifying each answer with a numerical score or other character symbol
- Identifying responses with codes is necessary if data is to be processed by computer

What does coding do?

- It serves as a rule for interpreting, classifying, and recording data
- It guides the establishment of category sets
 - Appropriate to the research problem and purpose
 - Exhaustive
 - Mutually exclusive
 - Derived from one classification principle

What is classification of data?

- It is the process of arranging data into homogeneous (similar) groups according to their common characteristics.
- Raw data cannot be easily understood, and it is not fit for further analysis and interpretation. Arrangement of data helps users in comparison and analysis.
- For example, the population of a town can be grouped according to sex, age, marital status, etc.

Objectives of Data Classification

The primary objectives of data classification are:

- To consolidate the volume of data in such a way that similarities and differences can be quickly understood. Figures can consequently be ordered in sections with common traits.
- To aid comparison.
- To point out the important characteristics of the data at a flash.
- To give importance to the prominent data collected while separating the optional elements.
- To allow a statistical method of the materials gathered.

Tabulation

Tabulation is a systematic and logical representation of numeric data in rows and columns to facilitate comparison and statistical analysis. It facilitates comparison by bringing related information close to each other and helps in statistical analysis and interpretation.

In other words, the method of placing organized data into a tabular form is known as tabulation. It may be complex, double, or simple, depending upon the nature of categorization.

Objectives Of Tabulation:

- **(1) To simplify complex data**
 - It reduces the bulk of information, i.e., it reduces raw data in a simplified and meaningful form so that it can be easily interpreted by a common man in less time.
- **(2) To bring out essential features of data**
 - It brings out the chief/main characteristics of data.
 - It presents facts clearly and precisely without textual explanation.
- **(3) To facilitate comparison**
 - The representation of data in rows and columns is helpful in simultaneous detailed comparison on the basis of several parameters.
- **(4) To facilitate statistical analysis**
 - Tables serve as the best source of organised data for statistical analysis.
 - The task of computing average, dispersion, correlation, etc., becomes easier if data is presented in the form of a table.
- **(5) To save space**
 - A table presents facts in a better way than the textual form.
 - It saves space without sacrificing the quality and quantity of data.

Rules to be followed while Tabulating

1. Every table should be numbered. Numbering could be in alphabet, Arabic or Roman or alpha numeric.
2. Each table should have a distinct title. The title should be short and clearly indicate what the contents of the table indicate.
3. Unit of measurement of the values in the table must be specified i.e., Rs. Crores, Million tonnes, etc.
4. Each column should be titled
5. Each row must be titled
6. Columns and rows are to be numbered
7. Any additional information about a cell [i.e. box] may be indicated using a symbol and giving the explanation of the symbol under the table.
8. Foot notes of the table should indicate the explanatory notes on the data in the table and the foot notes must be positioned below the table
9. Data to be compared must be placed in adjacent Columns
10. In case of numbers loaded in the cells, these numbers should be arranged as per digits.
11. The table should be simple and easily understandable.
12. Abbreviations should be avoided and if used, must be elaborated through foot notes.

Tabulate the following data taking 100-110, 110-120....
as the classes : **[05]**

138	164	115	150	120	102	140	150
115	125	140	147	125	142	115	103
142	155	115	125	163	119	135	111
125	107	145	165	125	130	142	135
160	119	109	110	146	166	116	136

Solution

Class Intervals	Tally Bars	Frequencies
100-110		5
110-120		8
120-130		7
130-140		6
140-150		8
150-160		2
160-170		4
Total		50

Prepare a Frequency Distribution Table for the following data :

The marks scored by 50 students in an examination paper are given below :

30 45 48 55 39 25 31 12 18 21 54 59 33
33 43 44 10 38 19 26 41 35 37 41 46 35
51 37 58 48 17 19 23 26 29 38 57 36
44 43 27 31 43 22 31 47 34 18 15 51

(Take classes as 10-20, 20-30,...) **[05]**

What is data validation?

Data validation is the process of examining the quality and accuracy of the collected data before processing and analysing it. It not only ensures the accuracy but also confirms the completeness of your data.

Objectives of data validation.

1. Error free data.
2. Ensuring that the appropriate data has been collected
3. Accuracy in analysis.
4. To find out feasibility of using statistical tools.

Analysis and Interpretation of data

Data Analysis

"Data analysis is the process of bringing order, structure and meaning to the mass of collected data. It is a messy, ambiguous, time-consuming, creative, and fascinating process. It does not proceed in a linear fashion; it is not neat".

Data analysis is the process of collecting, modeling, and analyzing data using various statistical and logical methods and techniques. Businesses rely on analytics processes and tools to extract insights that support strategic and operational decision-making.

- All these various methods are largely based on two core areas: *quantitative* and *qualitative research*.

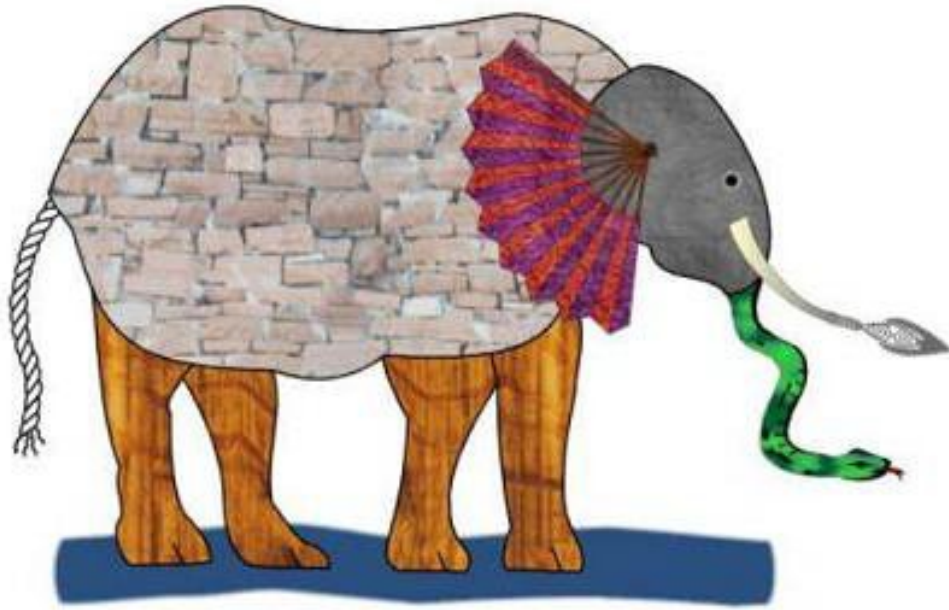
WHY DO WE ANALYZE DATA?

The purpose of analyzing data is to obtain usable and useful information. The analysis, irrespective of whether the data is qualitative or quantitative, may:

- Describe and summaries the data
- Identify relationships between variables
- Compare variables
- Identify the difference between variables
- Forecast outcomes

Process involved in Data analysis.





Blind men and an elephant
- Indian fable

Things aren't always what we think!

Six blind men go to observe an elephant. One feels the side and thinks the elephant is like a wall. One feels the tusk and thinks the elephant is like a spear. One touches the squirming trunk and thinks the elephant is like a snake. One feels the knee and thinks the elephant is like a tree. One touches the ear, and thinks the elephant is like a fan. One grasps the tail and thinks it is like a rope. They argue long and loud and though each was partly in the right, all were in the wrong.

Methods of Data Analysis

1. Based on nature of data:

- A. Qualitative Analysis: Qualitative data analysis (QDA) is the process of organizing, analyzing, and interpreting qualitative data—non-numeric, conceptual information and user feedback—to capture themes and patterns, answer research questions, and identify actions to take to improve your product.
- B. Quantitative analysis: Data Analysis can be explained as the process of discovering useful information by evaluating data whereas quantitative data analysis can be defined as the process of analyzing data that is number-based or data that can easily be converted into numbers.

2. Based on Purpose of data analysis

- C. Descriptive analysis: Descriptive analytics is the process of using current and historical data to identify trends and relationships. It's sometimes called the simplest form of data analysis because it describes trends and relationships but doesn't dig deeper.
- D. Predictive analysis: The term predictive analytics refers to the use **of statistics and modeling techniques to make predictions about future outcomes and** performance.
- E. Prescriptive analysis: It is a decision making activity based on data analysis. This analysis not only describe but also prescribe the decision to be taken.

Qualitative analysis Methods

1. Content analysis: Content analysis is a research method that examines and quantifies the presence of certain words, subjects, and concepts in text, image, video, or audio messages. The method transforms qualitative input into quantitative data to help you make reliable conclusions about what customers think of your brand, and how you can improve their experience and opinion.
2. Thematic analysis: Thematic analysis helps to identify, analyze, and interpret patterns in qualitative data, and can be done with tools
3. Narrative analysis: Narrative analysis is a method used to interpret research participants' stories—things like testimonials, case studies, interviews, and other text or visual data
4. Grounded theory analysis: Grounded theory analysis is a method of conducting qualitative research to develop theories by examining real-world data. The technique involves the creation of hypotheses and theories through the collection and evaluation of qualitative data, and can be performed with tools
5. Discourse analysis: It involves the observation of texts, audio, and videos to study the relationships between the information and its context.

Quantitative data analysis tools

- **Relate measurement scales with variables:** Associate measurement scales such as Nominal, Ordinal, Interval and Ratio with the variables. This step is important to arrange the data in proper order. Data can be entered into an excel sheet to organize it in a specific format.
- **Connect descriptive statistics with data:** Link descriptive statistics to encapsulate available data. It can be difficult to establish a pattern in the raw data. Some widely used descriptive statistics are:
 - Mean- An average of values for a specific variable
 - Median- A midpoint of the value scale for a variable
 - Mode- For a variable, the most common value
 - Frequency- Number of times a particular value is observed in the scale
 - Minimum and Maximum Values- Lowest and highest values for a scale
 - Percentages- Format to express scores and set of values for variables

Which calculation do I use? It depends upon what you want to know.

Do you want to know how many individuals checked each answer?	Frequency
Do you want the proportion of people who answered in a certain way?	Percentage
Do you want the average number or average score?	Mean
Do you want the middle value in a range of values or scores?	Median
Do you want to show the range in answers or scores?	Range
Do you want to compare one group to another?	Cross tab
Do you want to report changes from pre to post?	Change score
Do you want to show the degree to which a response varies from the mean?	Standard deviation

Data Interpretation:

Numbers do not speak for themselves.

For example, what does it mean that 55 youth reported a change in behavior. Or, 25% of participants rated the program a 5 and 75% rated it a 4. What do these numbers mean?

Interpretation is the process of attaching meaning to the data.

Interpretation demands fair and careful judgments. Often the same data can be interpreted in different ways. So, it is helpful to involve others or take time to hear how different people interpret the same information.

Think of ways you might do this...for example, hold a meeting with key stakeholders to discuss the data; ask individual participants what they think

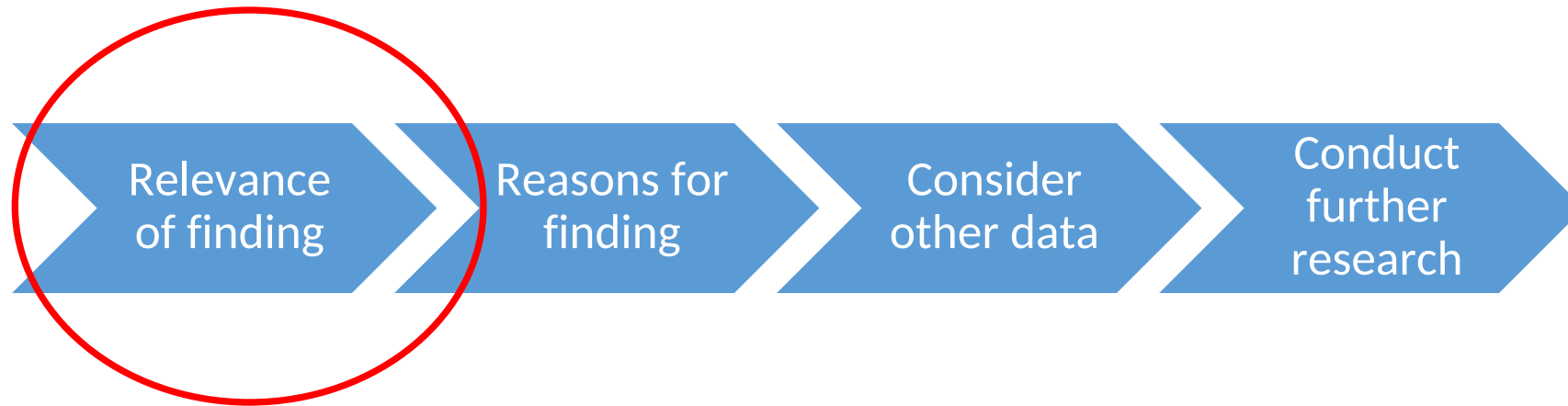
Process in Interpreting data

- Adding meaning to information by making connections and comparisons and exploring causes and consequences



Interpretation – relevance of finding

- Adding meaning to information by making connections and comparisons and exploring causes and consequences

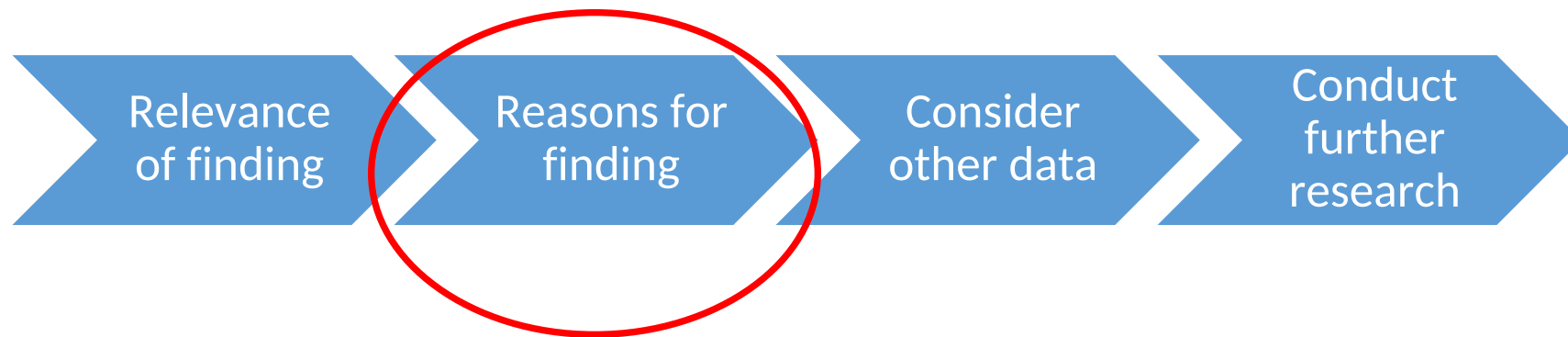


Interpretation – relevance of finding

- Does the indicator meet the target?
- How far from the target is it?
- How does it compare (to other time periods, other facilities)?
- Are there any extreme highs and lows in the data?

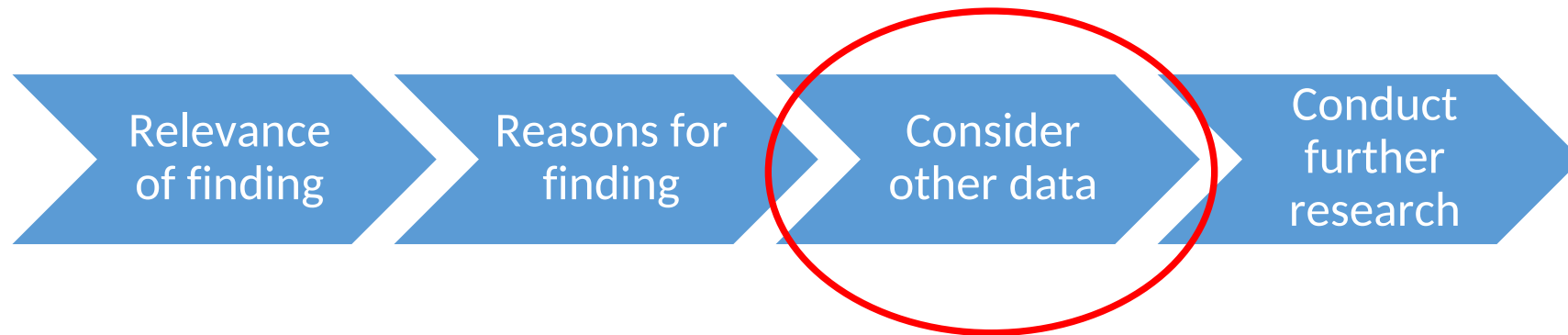
Interpretation – possible causes?

- Supplement with expert opinion
 - Others with knowledge of the program or target population



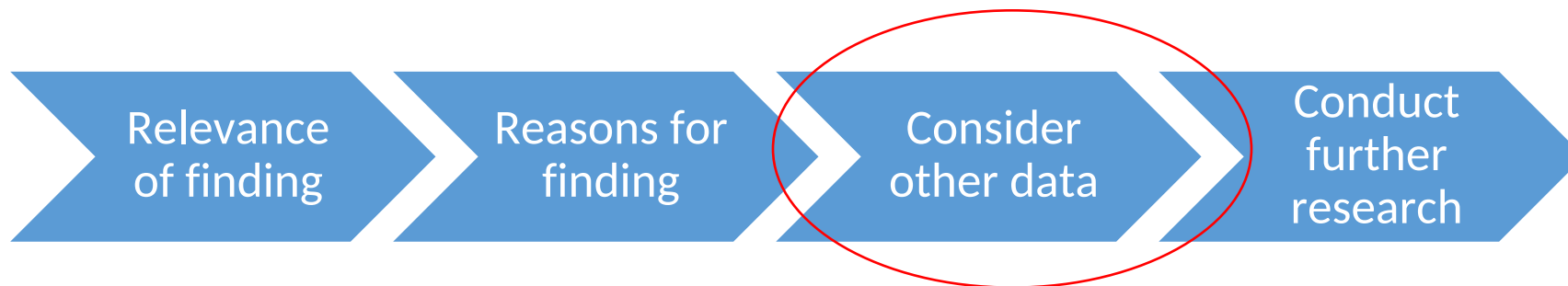
Interpretation – consider other data

Cross verify the other data set which have any specific meaning in the research and also include them in data interpretation. Express the limitations if any in generalizing the data interpretation.



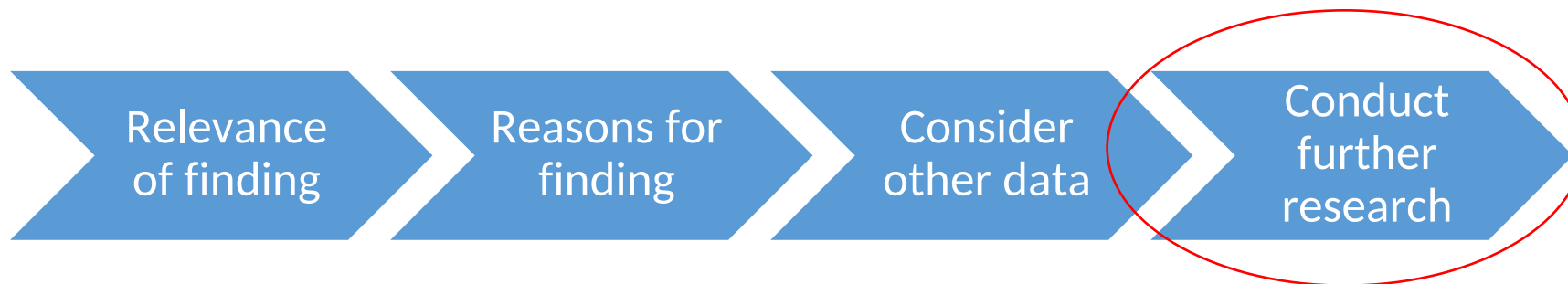
Interpretation – other data sources

- Situation analyses
- Demographic and health surveys
- Performance improvement data



Interpretation – conduct further research

- Data gap → conduct further research
- Methodology depends on questions being asked and resources available



Presentation of data in the report

The presentation of the data i.e. information should be presented as per understanding convenience of the report readers. The data presentation should pay attention to the following.

1. Easy Understanding
2. Should not be complex and time consuming in understanding the data.
3. Sequence or logical method be applied in presentation of data.
4. Reader is not much interested in calculations of data but only needs inferences and evidences for clarifications.

Report writing and Presentation of Data

- After data analysis and interpretation of data the next step is to present the data in the form of detailed report.
- Report writing is a formal style of writing elaborately on a topic. The tone of a report and report writing format is always formal. The important section to focus on is the target audience. For example – report writing about a school event, report writing about a business case, etc.

Effective way of presenting data

The complex statistical data may be presented in the following way

1. Tables
2. Graphs.
3. Pictorial presentations.
4. Use of colors and shapes to compare data.
5. Use of data visualization software tools

Important data visualization tools

Data visualization tools

From sources across the web

 Tableau	 Qlik	 Power BI
 Sisense	 D3.js	 Wrapper
 Google Charts	 Plotly	 FusionCharts
 Zoho Corporation	 Infogram	 Domo
 Highcharts	 Qlik Sense	 Google Data Studio
 Looker	 Leaflet	 RAWGraphs
 Tableau Public	 Chart.js	 ChartBlocks
 Grafana	 Klipfolio Inc.	 Microsoft Excel

Preparation of data in Graphs

What is Graph?

A graph can be defined as a pictorial representation or a diagram that represents data or values in an organized manner. The points on the graph often represent the relationship between two or more things.

Types of Graph:

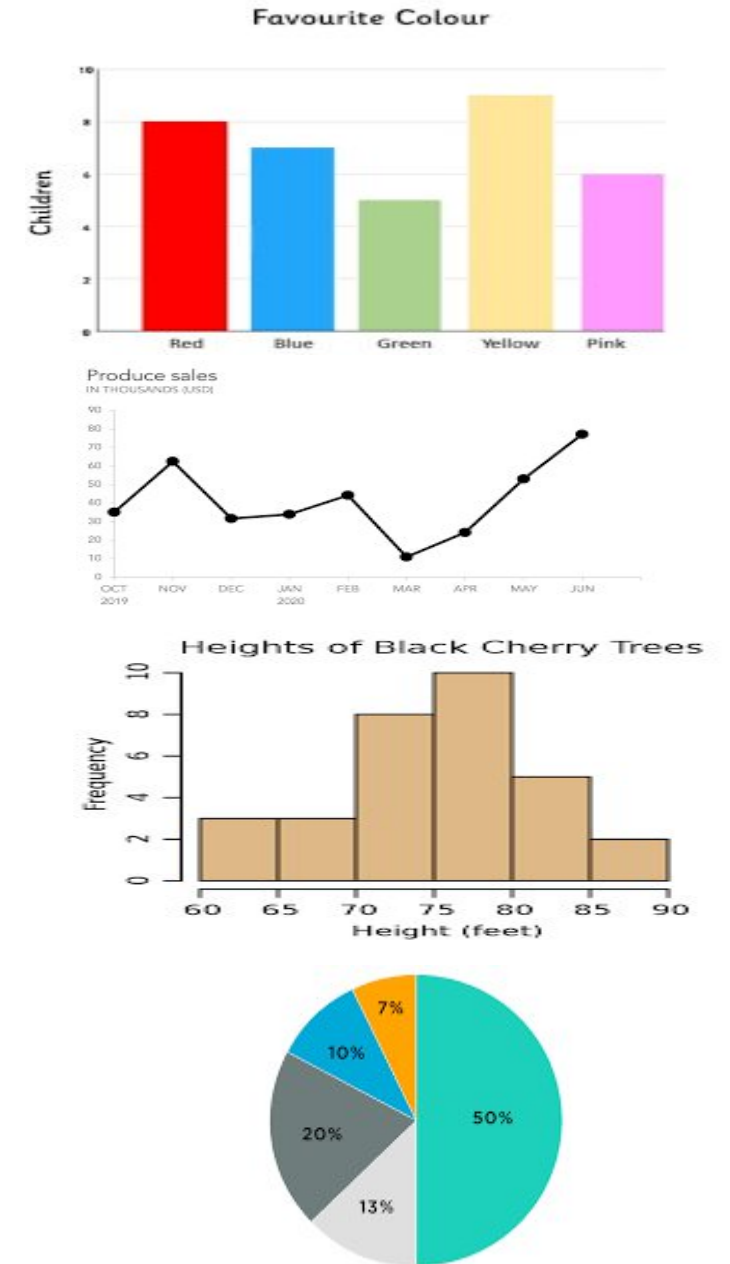
- Statistical graphs: Presenting statistics in graph
- Exponential graphs: Explaining exponential growth or catheter (Demand and supply)
- Logarithmic graphs: Expressed with mathematical equation.
- Trigonometric graphs: Graphs of trigonometric function

Statistical Graphs

A statistical graph or chart is defined as the pictorial representation of statistical data in graphical form. The statistical graphs are used to represent a set of data to make it easier to understand and interpret statistical information. The different types of graphs that are commonly used in statistics are given below.

Types of Statistical Graphs:

- Bar graph
- Line graph
- Histogram
- Pie chart



Importance of report writing: Written reports are evidences for;

- ▶ Monitor and Control operations
- ▶ Implement policies and procedures
- ▶ Comply with regulatory framework
- ▶ Documentation
- ▶ Analyze information and make decisions
- ▶ Obtain new business and venture into new areas
- ▶ Maintain contact with stakeholders

Research Reports

Discussion points

- Types of research reports
- Report structure
- Guidelines for effective documentation.

What is Research Report?

- A research report is a well-crafted document that outlines the processes, data, and findings of a systematic investigation. It is an important document that serves as a first-hand account of the research process, and it is typically considered an objective and accurate source of information.
- In many ways, a research report can be considered as a summary of the research process that clearly highlights findings, recommendations, and other important details. Reading a well-written research report should provide you with all the information you need about the core areas of the research process.

Types of Research Reports

Research reports vary greatly in length and type. In each individual case, both the length and the form are largely dictated by the problems at hand.

For instance, business firms prefer reports in the letter form, just one or two pages in length.

Banks, insurance organisations and financial institutions are generally fond of the short balance-sheet type of tabulation for their annual reports to their customers and shareholders.

Mathematicians prefer to write the results of their investigations in the form of algebraic notations.

Students of literature usually write long reports presenting the critical analysis of some writer or period or the like with a liberal use of quotations from the works of the author under discussion.

In the field of education and psychology, the favourite form is the report on the results of experimentation accompanied by the detailed statistical tabulations.

Clinical psychologists and social pathologists frequently find it necessary to make use of the case-history form.

Book-reviews which analyze the content of the book and report on the author's intentions, his success or failure in achieving his aims, his language, his style, scholarship, bias or his point of view. Such reviews also happen to be a kind of short report.

The reports prepared by governmental bureaus, special commissions, and similar other organisations are generally very comprehensive reports on the issues involved.

Similarly, Ph.D. theses and dissertations are also a form of report-writing, usually completed by students in academic institutions.

Types of Research Reports

1. Technical Research Reports.
2. Popular Research Reports.
3. Informational Research Reports.
4. Analytical Research Reports.
5. Proposal Reports (Report prepared for funding agencies)
6. Oral Research or Written Report Presentations
7. Formal or Informal reports.
8. Short or Long Reports.
9. Internal or External Research Reports.
10. Periodic Research reports

Technical Report

In the technical report the main emphasis is on

- (i) the methods employed,
- (it) assumptions made in the course of the study,
- (iii) the detailed presentation of the findings including their limitations and supporting data.

A general outline of a technical report can be as follows:

1. **Summary of results:** A brief review of the main findings just in two or three pages.
2. **Nature of the study:** Description of the general objectives of study, formulation of the problem in operational terms, the working hypothesis, the type of analysis and data required, etc.
3. **Methods employed:** Specific methods used in the study and their limitations. For instance, in sampling studies we should give details of sample design viz., sample size, sample selection, etc.

4. **Data:** Discussion of data collected, their sources, characteristics and limitations. If secondary data are used, their suitability to the problem at hand be fully assessed. In case of a survey, the manner in which data were collected should be fully described.
5. **Analysis of data and presentation of findings:** The analysis of data and presentation of the findings of the study with supporting data in the form of tables and charts be fully narrated. This, in fact, happens to be the main body of the report usually extending over several chapters.
6. **Conclusions:** A detailed summary of the findings and the policy implications drawn from the results be explained.
7. **Bibliography:** Bibliography of various sources consulted be prepared and attached.
8. **Technical appendices:** Appendices be given for all technical matters relating to questionnaire, mathematical derivations, elaboration on particular technique of analysis and the like ones.
9. **Index:** Index must be prepared and be given invariably in the report at the end.

Popular report

The popular report is one which gives emphasis on simplicity and attractiveness.

The simplification should be sought through clear writing, minimization of technical, particularly mathematical, details and liberal use of charts and diagrams.

Attractive layout along with large print, many subheadings, even an occasional cartoon now and then is another characteristic feature of the popular report.

Besides, in such a report emphasis is given on practical aspects and policy implications.

Features of a popular report

1. **The findings and their implications:** Emphasis in the report is given on the findings of most practical interest and on the implications of these findings.
2. **Recommendations for action:** Recommendations for action on the basis of the findings of the study is made in this section of the report.
3. **Objective of the study:** A general review of how the problem arise is presented along with the specific objectives of the project under study.
4. **Methods employed:** A brief and non-technical description of the methods and techniques used, including a short review of the data on which the study is based, is given in this part of the report.
5. **Results:** This section constitutes the main body of the report wherein the results of the study are presented in clear and non-technical terms with liberal use of all sorts of illustrations such as charts, diagrams and the like ones.
6. **Technical appendices:** More detailed information on methods used, forms, etc. is presented in the form of appendices. But the appendices are often not detailed if the report is entirely meant for general public.

Oral Presentation

At times oral presentation of the results of the study is considered effective, particularly in cases where policy recommendations are indicated by project results.

The merit of this approach lies in the fact that it provides an opportunity for give-and-take decisions which generally lead to a better understanding of the findings and their implications.

But the main demerit of this sort of presentation is the lack of any permanent record concerning the research details and it may be just possible that the findings may fade away from people's memory even before an action is taken.

In order to overcome this difficulty, a written report may be circulated before the oral presentation and referred to frequently during the discussion.

Oral presentation is effective when supplemented by various visual devices.

Written Reports

Normally most of the research reports are presented in the form of written reports. The written reports play an important role in business and have advantages over oral reports.

The written report enables the manager to weigh the facts and arguments presented therein, and implement the acceptable recommendations, with a view to closing the gap between the existing state of affairs and the desired state.

Principles of written reports

The written report should follow SIMPLE principle.

- **SHORT** – Only relevant and important aspect of research standpoint. Everything is briefly stated.
- **INTERESTING** – The style, format and contents of a report must attract and maintain the attention of the readers.
- **METHODICAL** – The sections of a report and their contents must be selected with the readers in mind and arranged in a logical sequence. The format of contents for a report given in the chapter should be used as a guidance.
- **PRECISE** – Report must be clear & comprehensive. Comprehensive implies that the report must contain enough details to provide an accurate, complete picture of the study.
- **LUCID** – Use simple words and sentences. Appropriate Graphical illustration, tables and charts used to communicate contents clearly and quickly .
- **ERROR FREE** – Minor arithmetical, grammatical or typographical errors may not ensure credibility of the report.

Structure of Research Report

TYPICAL RESEARCH REPORT STRUCTURE

1. Title Page
2. Abstract
3. Acknowledgements (optional)
4. Contents
 - List of Tables
 - List of Figures
5. Introduction
6. Literature Review
7. Methodology
8. Results
9. Discussion
10. Conclusions + Recommendations
11. Reference List
12. Appendices

Guidelines for effective documentations in Research

What Is Documentation?

- Is acknowledging sources that we have used in our research

What should be documented?

- Research documentation shall include the intellectual and practical aspects of the research, and, when necessary, refer to the administrative documents, such as funding, permits and agreements, relating to the research.

For ongoing research, the following must be documented:

- Background and purpose
- Materials and methods (includes even processing and analysis of data)
- Results (raw and processed data)
- Conclusions

Why is documentation important?

1. Documentation informs readers that you have consulted experts on the subject, and have relied on the most current and authoritative sources to support your statements; -
2. Gives proper credit to the sources used in the argument, which is an ethical requirement. Failure to do so will result in being accused of plagiarism; -
3. Provides readers the books and articles you have used in support of your argument(s). As a result, the readers may be encouraged to go to the sources for additional information.

What must be documented?

1. Any direct quotation, even a single phrase or key word, must be identified according to the work, and the exact place in that work
2. Any paraphrase or summary of another individual's written work or from an oral report or presentation
3. Any opinion (verbal or writing) without which you could not have reached your views without the help of another source
4. Any statistical data which you have not compiled yourself
5. Any visuals that you have not prepared yourself
6. Any software programs that you did not develop yourself.

Documentation in writing process

Documentation begins as soon as you start researching, and it continues throughout the writing process: drafting, revising, and editing. Therefore, you need to maintain a careful record of sources you use and the exact material you take from them. You need to check the accuracy of your documentation during the various stages of the writing process.

Types of Documentations used in research

1. Parenthetical documentation (Mention of documents referred in research at the end of the report under references as an annexure)
2. Documentation through footnotes and endnotes. (Mentioned at the end of the page or in the footage of the page)

Guidelines for effective documentation

1. Fit for context (Relevance)
2. Clearly written and to the point
3. Visual where possible
4. Classified in systematic manner
5. Up to date and contemporary
6. Discoverable & Tracked

Thank You